

Yamaha Extended Control API Specification (Basic)

Rev. 2.50

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1. Preface

This document describes specifications of API of Yamaha Extended Control (YXC, hereafter) for Yamaha A/V products. YXC is Yamaha's new communication protocol sent over Ethernet and Wi-Fi to control MusicCast enabled devices. This document describes basic specifications including the way to control power, change input, browse list, retrieve album art. Advanced function including MusicCast link function and others is described in Yamaha Extended Control API Specification (Advanced).

When using YXC, there may be items in the telegram that are not described in this specification, but they do not affect the operation, so please disregard them.

This specification (Rev.2.50) applies only to MusicCast devices that have been upgraded to a product firmware that supports after API version 2.00.

2. Definitions

- MusicCast
A function to share music easily among audio devices in multiple rooms at homes
- MusicCast Device(s)
All products/devices that are equipped with MusicCast functions
- MusicCast Network
A communication network on which MusicCast Devices join.
 - By multiple MusicCast Devices communicating each other on the same network
(=MusicCast Network), all MusicCast devices can be operated using MusicCast App/YXC
- Location
A unit of MusicCast Network being comprised. Location is expressed as Location ID made up by 32 characters
 - Max number of MusicCast Devices within 1 Location is up to 32 (available on or after API version 2.00: the max number before version 2.00 was 10).
- Link
A function to share an audio stream among rooms
- Device
Equal to MusicCast Device
- Room
A minimum component MusicCast devices can be shown/associated within a MusicCast network

3. API Overview

- Request Parameters
Parameter information to inquire. Parameters stated as "required = yes" have always to be specified as one of inquiring parameters, while parameters marked as "required = no" can be omitted or need to be used in special cases.
- Response Parameters
Parameter information to be returned in the body of response (in JSON object format). Not all parameters listed here are always to be returned in a response, depending on each device specs/functions/setup situations. "response_code" is always returned.
 - Note: JSON format data expressions in this document get well formatted for readability, but actual data don't include spaces, intends and carriage return/line feeds
- Base URL
"<BaseURL>" shown in URI is <http://{host}/YamahaExtendedControl>
 - {host} is an IP address of a Device to get controlled
- API version
Values like "v1" or "v2" specified in URI shows a corresponding API version number. Backward compatibility of APIs always being assured unless specified, all APIs that have equal to or smaller API version number(s) that can be read via "getDeviceInfo" are supported on a Device.
- Data size of a sting
In case a string is specified with its data size, it is a size before JSON escape process

4. System

4.1. getDeviceInfo

For retrieving basic information of a Device.

Request Parameters

URI	<BaseURL>/v1/system/getDeviceInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getDeviceInfo

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details
model_name	string	Returns model name Ex. "RX-V479" / "WXC-50" / "UNKNOWN"
destination	string	Returns Device's destination region code. There are cases of multiple regions codes like "UC" "BG" depending on Devices Values: "J" / "U" / "C" / "R" / "B" / "G" / "BG" / "T" / "A" / "L" / "K" / "E" / "V" / "F" / "P" / "S" / "H"
device_id	string	Return Device's ID (12 digit ASCII).Device ID is unique ID to identify Device. Note: Available on and after API Version 1.17
system_id	string	Returns System ID (8 digit ASCII)
system_version	floating point	Returns System Version
api_version	floating point	Returns API Version
netmodule_generation	integer	Returns generation number of the network module (Available on after API version 1.20)
netmodule_version	string	Returns Network Module version
netmodule_checksum	string	Return check-sum of Network Module in hex. "00000000" is returned until check-sum calculation is done
serial_number	string	Returns Serial Number (16 alphanumeric characters)
category_code	integer	Returns category code of the model. If this parameter does not exist, treat it as 0 (reserved). Values: 0 : reserved 1 : AV Receiver 2 : Sound Bar 3 : Stereo Receiver 4 : Subwoofer 5 : Mini System 6 : Desktop Audio 1
net_module_num	integer	Returns number of network modules that the device has. This parameter exists only on devices with multiple modules (ex. XDA-QS5400).

Example Response	<pre>{ "response_code":0, "model_name":"RX-V679", "destination":"U", "system_id":"ABADCAFE", "system_version":1.10, "api_version":2.00, "netmodule_generation":2, "netmodule_version":"0200", "netmodule_checksum":"878059DD", "net_module_num":4}</pre>
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4.2. getFeatures

For retrieving feature information equipped with a Device.

Request Parameters

URI	<BaseURL>/v1/system/getFeatures
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getFeatures

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details
system	object	Returns system's overall info
- func_list	string array	Returns valid functions Values: "wired_lan" / "wireless_lan" / "wireless_direct" / "extend_1_band" / "dfs_option" / "network_standby" / "network_standby_auto" / "bluetooth_standby" / "bluetooth_tx_setting" / "bluetooth_tx_connectivity_type" / "auto_power_standby" / "ir_sensor" / "speaker_a" / "speaker_b" / "headphone" / "dimmer" / "zone_b_volume_sync" / "hdmi_out_1" / "hdmi_out_2" / "hdmi_out_3" / "airplay" / "stereo_pair" / "speaker_settings" / "disklavier_settings" / "background_download" / "remote_info" / "network_reboot" / "system_reboot" / "ypao_setting" / "ypao_speaker_unit" / "ypao_volume" / "auto_play" / "speaker_pattern" / "turn_table_rotation_time" / "party_mode" / "name_text_avr" / "video_preset" / "party_volume" / "party_mute" / "hdmi_standby_through" / "analytics"
- zone_num	integer	Returns Zone numbers. Zone B is treated as Zone2 in YXC so a Device with ZoneB returns 2. A Device without Zones returns 1
- input_list	array	Returns a list of inputs
- id	string	Returns Input ID Refer to "ALL ID List" for details
- distribution_enable	boolean	Returns whether an input can be a source of Link distribution
- rename_enable	boolean	Returns whether an input can be renamed

- account_enable	boolean	Return whether an input comes with an account info
- play_info_type	string	Returns a type of playback info. Depending on this type, use specific API to retrieve appropriate playback info Values: "none" / "tuner" / "netusb" / "cd"
- range_step	array	Returns minimum/maximum/step values of a parameter
- id	string	Returns an ID Value: "dimmer"
- min	integer	Returns a minimum value of a parameter
- max	integer	Returns a maximum value of a parameter
- step	integer	Returns a step value of a parameter
- bluetooth	object	Returns a information related to Bluetooth
- update_cancelable	boolean	Returns a whether updating can be canceled and connected to the device during Bluetooth (Sink) device list update
- speaker_pattern_num	integer	Returns the number of patterns that can be selected with the speaker setting function
- video_preset_num	integer	Returns the total number of selectable Video Presets. Models without Video Preset do not have this parameter.
- party_volume_list	string array	Returns the information to set the volume of the zone joining in the party mode at the same time. Value: "up" / "down"
- hdmi_standby_through_list	string array	Returns HDMI Standby Through settings. Value: "off" / "on" / "auto"
zone	array	Returns Zone related information
- id	string	Returns Zone ID. Refer to " ALL ID List " for details
- zone_b	boolean	Returns whether the target Zone is Zone B or not. Valid only when Zone ID is "zone2"
- func_list	string array	Returns a list of valid functions Values: "power" / "sleep" / "volume" / "mute" / "sound_program" / "surround_ai" / "surround_3d" / "direct" / "pure_direct" / "enhancer" / "tone_control" / "equalizer" / "balance" / "dialogue_level" / "dialogue_lift" / "clear_voice" / "subwoofer_volume" / "bass_extension" / "signal_info" / "prepare_input_change" / "link_control" / "link_audio_delay" / "link_audio_quality" / "scene" / "contents_display" / "cursor" / "menu" / "actual_volume" / "audio_select" / "surr_decoder_type" / "extra_bass" / "adaptive_drc" / "dts_dialogue_control" / "mono" / "adaptive_dsp_level"
- input_list	string array	Returns a list of selectable Input IDs Refer to " ALL ID List " for details
- sound_program_list	string array	Returns a list of selectable Sound Program IDs Refer to " ALL ID List " for details
- surr_decoder_type_list	string array	Returns a list of selectable Surround Decoder Types Values: "toggle" / "auto" / "dolby_pl" / "dolby_pl2x_movie" / "dolby_pl2x_music" / "dolby_pl2x_game" / "dolby_surround" / "dts_neural_x" / "dts_neo6_cinema" / "dts_neo6_music"
- tone_control_mode_list	string array	Returns selectable settings of Tone Control Mode. If there's no list of this, it's fixed to "manual"

		Values: "manual" / "auto" / "bypass"
- equalizer_mode_list	string array	Returns selectable settings of Equalizer Mode. If there's not list of this, it's fixed to "manual" Values: "manual" / "auto" / "bypass"
- link_control_list	string array	Returns selectable settings of Link Control Values: "standard" / "stability" / "speed"
- link_audio_delay_list	string array	Returns selectable settings of Link Audio Delay Values: "lip_sync"/"audio_sync" / "audio_sync_on" / "audio_sync_off" / "balanced"
- link_audio_quality_list	string array	Returns selectable settings of Link Audio Quality Values: "compressed" / "uncompressed"
- range_step	array	Returns ID, min, max, step values of a parameter
- id	string	Returns ID of a parameter Values: "volume" / "tone_control" / "equalizer" / "balance" / "dialogue_level" / "dialogue_lift" / "subwoofer_volume" / "actual_volume_db" / "actual_volume_numeric" / "dts_dialogue_control"
- min	floating point	Returns minimum value of a parameter * Please be aware that the type has been changed from "integer" to "floating point" in API Version 2.00 or later. * There is a possibility that it may be stored as "floating point" only when the ID is "actual_volume_db" or "actual_volume_numeric". For other IDs, they are stored as "integer".
- max	floating point	Returns maximum value of a parameter * Please be aware that the type has been changed from "integer" to "floating point" in API Version 2.00 or later. * There is a possibility that it may be stored as "floating point" only when the ID is "actual_volume_db" or "actual_volume_numeric". For other IDs, they are stored as "integer".
- step	floating point	Returns step value of a parameter * Please be aware that the type has been changed from "integer" to "floating point" in API Version 2.00 or later. * There is a possibility that it may be stored as "floating point" only when the ID is "actual_volume_db" or "actual_volume_numeric". For other IDs, they are stored as "integer".
- scene_num	integer	Returns selectable scenes of total number This parameter does not exist in models without a scene
- cursor_list	string array	Returns functions available with executeCursor of remote control function This parameter does not exist in models without a function Values: "up" / "down" / "left" / "right" / "select" / "return"
- menu_list	string array	Returns functions available with executeMenu of remote control function This parameter does not exist in models without a function Values: "on_screen" / "top_menu" / "menu" / "option" / "display" / "help" / "home" / "mode" / "red" / "green" / "yellow" / "blue"

- actual_volume_mode_list	string array	Returns function list of configurable Actual Volume Mode This parameter does not exist in models without a function Values: "db" / "numeric" Note: For models fixed to "db", the value is "db" only.
- audio_select_list	string array	Returns function list of configurable Audio Select This parameter does not exist in models without a function Values: "auto" / "hdmi" / "coax_opt" / "analog" / "unavailable"
- ccs_supported	string array	Returns a list of features available in Cloud Connect Service (CCS). Values: "connected_speaker"
tuner	object	Returns information related to Tuner function
- func_list	string array	Returns a list of available functions Values: "am" / "fm" / "rds" / "dab" / "hd_radio" / "fm_auto_preset" / "dab_initial_scan" / "dab_tune_aid"
- range_step	array	Returns object ID, min, max, step values of a parameter
- id	string	Returns object ID of a parameter Values: "am" / "fm"
- min	integer	Returns minimum value (in unit of kHz)
- max	integer	Returns maximum value (in unit of kHz)
- step	integer	Returns step value (in unit of kHz)
- preset	object	Returns information related to Preset
- type	string	Returns Preset type. The value of "common" is for each band. Values: "common" (band common) / "separate" (each band preset)
- num	integer	Returns preset capable number. If type is "separate", each band of AM/FM/DAB can have presets up to this number
netusb	object	Returns information related to Net/USB function
- func_list	string array	Returns a list of available functions Value: "recent_info" / "play_queue" / "mc_playlist" / "streaming_service_use"
- preset	object	Returns information related to Preset
- num	integer	Returns preset capable number. All sources of Net/USB Input share these presets
- recent_info	object	Returns information related to playback history
- num	integer	Returns number of memorable playback history. Playback history is shared among all Net/USB Input sources
- net_radio_type	string	Returns type of Net Radio. If there's no list of this, it's fixed to "vTuner" Value : "vtuner" / "airable"
- pandora	object	Returns information related to Pandora
- sort_option_list	string array	Returns a list of selectable sorting method Value : "date" : creation date order (deprecated) "alphabet" : alphabet order "recent" : playback date order
- siriusxm	object	Returns information related to SiriusXM
- api_type	string	Returns API type. If this parameter does not exist, it is determined as "v2". Value: "v2" / "everest"
distribution	object	Returns information related to distribution function

- version	floating point	Returns version of distribution function Note : If devices of the same major version are used, it is guaranteed that the distribution function operates normally If this parameter does not exist, the version is handled as 1.xx (Major version = 1)
- compatible_client	integer array	Returns major version of compatible client delivery system when it works as a distribution server Note : Only devices that have versions listed here are certified as distribution clients and if this parameter does not exist, it is compatible only with Major version 1 client
- client_max	integer	Returns maximum number of devices that can be specified as clients when it works as a distribution server Note : If this parameter does not exist, the maximum number is 9
- server_zone_list	string array	Returns a list of Zone ID that can work as a sender(master) of distributing server. If a Device doesn't have anything of this parameter, "main" is the one of Zone ID to be able to work as a sender(master) Values: "main" / "zone2" / "zone3" / "zone4"
- mc_surround	object	Returns information related to MusicCast Surround function
- version	floating point	Returns MusicCast Surround System version The MusicCast Surround function is guaranteed to work properly for devices with the same Major version.
- func_list	string array	Returns a list of valid functions Value: "mc_speaker_settings" / "mcscs_error_info"
- master_role	object	Returns information related to role of sender(master) device
- surround_pair	boolean	Returns whether this device can operate as sender(master) of surround pair
- stereo_pair	boolean	Returns whether this device can operate as sender(master) of stereo pair. When it operates as a sender(master) device of stereo pair, it can also operate as a receiver(slave) device.
- subwoofer_pair	boolean	Returns whether this device can operate as sender(master) of subwoofer pair
- slave_role	object	Returns information related to role of receiver(slave) device
- surround_pair_l_or_r	boolean	Returns whether or not it can operate as a L or R monoral speaker of receiver(slave) device of surround pair
- surround_pair_lr	boolean	Returns whether or not it can operate as a LR stereo speaker of receiver(slave) device of surround pair
- subwoofer_pair	boolean	Returns whether this device can operate as receiver(slave) of subwoofer pair
clock	object	Returns information related to clock function
- func_list	string array	Returns a list of valid functions Values: "date_and_time" / "alarm" / "snooze" / "format"
- range_step	array	Returns minimum/maximum/step values of a parameter
- id	string	Returns ID of a parameter Values: "alarm_volume" / "alarm_fade"
- min	integer	Returns minimum value of a parameter
- max	integer	Returns maximum value of a parameter

- step	integer	Returns step value of a parameter
- alarm_fade_type_num	integer	Returns alarm fade type numbers
- alarm_mode_list	string array	Returns a list of selectable alarm mode Values: "oneday" / "weekly"
- alarm_input_list	string array	Returns a list of selectable Input IDs when resume playback Refer to " ALL ID List " for details
- alarm_preset_list	string array	Returns a list of selectable type when preset playback Values: "netusb" / "tuner_common" / "am" / "fm" / "dab"

Example Response	<pre>{ "response_code":0, "system":{ "func_list":["wired_lan", "wireless_lan", "wireless_direct", "network_standby", "bluetooth_standby", "speaker_settings", "auto_play", "speaker_pattern", "party_mode", "name_text_avr", "video_preset", "party_volume", "party_mute", "hdmi_standby_through"], "zone_num":3, "input_list":[{ "id":"hdmi1", "distribution_enable":false, "rename_enable":true, "account_enable":false }, { "id":"hdmi2", "distribution_enable":false, "rename_enable":true, "account_enable":false }, ... { "id":"spotify", "distribution_enable":true, "rename_enable":false, "account_enable":false }] } }</pre>
------------------	--

```

],

"speaker_pattern_num":2,
"video_preset_num":6,
"party_volume_list":[
  "up",
  "down"
],
"hdmi_standby_through_list":[
  "off",
  "on",
  "auto"
],
},
"zone":[
  {
    "id":"main",
    "func_list":[
      "power",
      "sleep",
      "volume",
      "mute",
      "sound_program",
      "surround_ai",
      "surround_3d",
      "direct",
      "pure_direct",
      "enhancer",
      "tone_control",
      "prepare_input_change",
      "link_control",
      "scene",
      "contents_display",
      "cursor",
      "menu",
      "actual_volume",
      "audio_select",
      "surr_decoder_type",
      "extra_bass",
      "adaptive_drc",
      "dts_dialogue_control",
      "mono",
      "adaptive_dsp_level"
    ],
    "input_list":[
      "hdmi1",
      "hdmi2",
      ...
      "spotify"
    ]
  }
]

```

```

],
"sound_program_list":[
  "munich",
  "vienna",
  ...
  "straight"
],
"surr_decoder_type_list":[
  "auto",
  "dolby_pl",
  ...
  "dts_neo6_music"
],
"link_control_list":[
  "standard",
  "stability"
],
"range_step":[
  {
    "id":"volume",
    "min":0,
    "max":194,
    "step":1
  },
  {
    "id":"tone_control",
    "min":-12,
    "max":12,
    "step":1
  }
  {
    "id":"actual_volume_db",
    "min":-80.5
    "max":16.5
    "step":0.5
  },
  {
    "id":"actual_volume_numeric",
    "min":0.0
    "max":97.0
    "step":0.5
  },
  {
    "id":"dts_dialogue_control",
    "min":0,
    "max":6,
    "step":1
  }
],

```



```

"scene_num":8,
"cursor_list":[
  "up",
  "down",
  "left",
  "right",
  "select",
  "return",
],
"menu_list":[
  "on_screen",
  "top_menu",
  "menu",
  "option",
  "display",
  "help",
  "home",
  "mode",
  "red",
  "green",
  "yellow",
  "blue"
],
"actual_volume_mode_list":[
  "db",
  "numeric"
],
"audio_select_list":[
  "auto",
  "hdmi",
  "coax_opt",
  "analog",
  "unavailable"
]
},
{
  "id":"zone2",
  "zone_b":false,
  "func_list":[
    "power",
    "sleep",
    "volume",
    "mute",
    "tone_control",
    "prepare_input_change"
  ],
  "input_list":[
    "hdmi1",
    "hdmi2",

```

```

...
"spotify"
],
"range_step":[
{
"id":"volume",
"min":0,
"max":194,
"step":1
},
{
"id":"tone_control",
"min":-5,
"max":5,
"step":1
}
{
"id":"actual_volume_db",
"min":-80.5
"max":16.5
"step":0.5
},
{
"id":"actual_volume_numeric",
"min":0.0
"max":97.0
"step":0.5
}
],
"scene_num":8,
"cursor_list":[
"up",
"down",
"left",
"right",
"select",
"return"
],
"menu_list":[
"on_screen",
"top_menu",
"menu",
"option",
"display",
"help",
"home",
"mode",
"red",
"green",

```

	<pre> "yellow", "blue"], "actual_volume_mode_list":["db", "numeric"] }], "tuner":{ "func_list":["am", "fm", "rds", "dab"], "range_step":[{ "id":"am", "min":531, "max":1611, "step":9 }, { "id":"fm", "min":76000, "max":90000, "step":100 }], "preset":{ "type":"common", "num":40 } }, "netusb":{ "preset":{ "num":40 }, "pandora":{ "sort_option_list":["recent", "alphabet"] } }, "distribution":{ "version":3.10, "compatible_client":[2, </pre>
--	--

```

    3
  ],
  "client_max":9,
  "server_zone_list":[
    "main"
  ],
  "mc_surround":{
    "master_role":{
      "surround_pair":true,
      "stereo_pair":false,
      "subwoofer_pair":true
    },
    "slave_role":{
      "surround_pair_l_or_r":false,
      "surround_pair_lr":false,
      "subwoofer_pair":false
    }
  }
}
}

```

4.3. getNetworkStatus

For retrieving network related setup / information.

Request Parameters

URI	<BaseURL>/v1/system/getNetworkStatus
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getNetworkStatus

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details
network_name	String	Returns Network Name (Friendly Name)
connection	String	Returns setup status of Connection Values: "wired_lan" / "wireless_lan" / "wireless_direct" / "extend_1" / "extend_2" / "extend_3" / "unknown"
dhcp	Boolean	Returns DHCP setup
ip_address	String	Returns IP Address
subnet_mask	String	Returns Subnet Mask
default_gateway	String	Returns Default Gateway
dns_server_1	String	Returns DNS Server 1
dns_server_2	String	Returns DNS Server 2
wireless_lan	Object	Returns information related to wireless network (Wi-Fi)
- enable	Boolean	Returns whether Wireless LAN is enabled or not. If it is disabled, you cannot use setWirelessLan to configure the wireless network. If this parameter does not exist, treat the value as if it were set to “true”.
- ssid	String	Returns SSID

- type	String	Returns encryption type Values: "none" / "wep" / "wpa2-psk(aes)" / "mixed_mode"
- ch	Integer	Returns a channel number of connected AP. Returns 0 when disconnected
- strength	Integer	Returns wireless strength. Returns -1 when Link is down status Value Range: -1 (Link Down)、 0 ~ 100
wireless_direct	Object	Returns information related to wireless network (Wireless Direct)
- ssid	String	Returns SSID
- type	String	Returns encryption type Values: "none" / "wpa2-psk(aes)"
musiccast_network	Object	Returns information related to MusicCast Network
- ready	Boolean	Returns a status whether or not a Device joins to MusicCast Network
- device_type	String	Returns a type of MusicCast Network on a Device. Returns "unknown" If not joined to MusicCast Network yet Values: "root" / "node" / "leaf" / "standard" / "unknown"
- child_num	Integer	Returns clients number connected to a Device
- ch	Integer	Returns a channel number that a Device discloses. Returns 0 when device_type is other than root or node
- initial_join_running	Boolean	Returns whether or not Initial Join is being executed
mac_address	Object	Returns MAC Address (12 digit ASCII)
- wired_lan	String	Returns wired MAC Address
- wireless_lan	String	Returns wireless (Wi-Fi) MAC Address
- wireless_direct	String	Returns wireless (Wireless Direct) MAC Address
airplay_pin	String	Returns AirPlay PIN
ipv6	Object	Returns information related to IPv6
- enable	Boolean	Returns IPv6 settings
- address	String	Returns IPv6 Address. This parameter does not exist if enable is false.
each_module_ip_list	string array	Returns IP Address information of each network module installed in the device. This parameter exists only on devices with multiple modules (ex. XDA-QS5400).

Example Response	<pre>{ "response_code":0, "network_name":"Yamaha AVR", "connection":"wired_lan", "dhcp":true, "ip_address":"192.168.0.10", "subnet_mask":"255.255.255.0", "default_gateway":"192.168.0.1", "dns_server_1":"192.168.0.1", "dns_server_2":"192.168.0.2", "wireless_lan":{ "enable": true, "ssid": "", "type": "none", "key": "", "ch": 0, </pre>
------------------	--

	<pre> "strength":0 }, "musiccast_network":{ "ready":true, "device_type":"root", "child_num":1, "initial_join_running":false }, "mac_address":{ "wired_lan":"00A0DE012345", "wireless_lan":"78A501012345", "wireless_direct":"78A501012346" }, "airplay_pin":"abc123", "ipv6":{ "enable":true, "address":"fe80::fedc:ba98:7654:3210" }, "each_module_ip_list":{ "192.168.0.10", ... "192.168.0.20" } } </pre>
--	--

4.4. setWiredLan

For setting Wired Network. Network connection is switched to wired by using this API. If no parameter is specified, current parameter is used. If set parameter is incomplete, it is possible not to provide network availability.

Request Parameters

name	Type	required	description
dhcp	String	no	Specifies DHCP setting
ip_address	String	no	Specifies IP Address
subnet_mask	String	no	Specifies Subnet Mask
default_gateway	String	no	Specifies Default Gateway
dns_server_1	String	no	Specifies DNS Server 1
dns_server_2	String	no	Specifies DNS Server 2

URI	<BaseURL>/v1/system/setWiredLan
Method	POST
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setWiredLan
Example Request (json)	<pre> { "dhcp":false, "ip_address":"192.168.0.11", "subnet_mask":"255.255.255.0", "default_gateway":"192.168.0.1", </pre>

	<pre> "dns_server_1":"192.168.0.1", "dns_server_2":"0.0.0.0" } </pre>
--	---

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details

Example Response	<pre> { "response_code":0 } </pre>
------------------	--------------------------------------

4.5. setWirelessLan

For setting Wireless Network (Wi-Fi). Network connection is switched to wireless (Wi-Fi) by using this API. If no parameter is specified, current parameter is used. If set parameter is incomplete, it is possible not to provide network availability.

Request Parameters

name	type	required	description
ssid	string	no	Specifies access point SSID (UTF-8 within 32 bytes)
type	string	no	Specifies wireless encryption protocol Values: "none" / "wep" / "wpa2-psk(aes)" / "mixed_mode"
key	string	no	Specifies encryption key (printable ASCII (0x20-0x7E) within 64 characters). This value is invalid when type "none" is specified.
dhcp	string	no	Specifies DHCP setting
ip_address	string	no	Specifies IP Address
subnet_mask	string	no	Specifies Subnet Mask
default_gateway	string	no	Specifies Default Gateway
dns_server_1	string	no	Specifies DNS Server 1
dns_server_2	string	no	Specifies DNS Server 2

URI	<BaseURL>/v1/system/setWirelessLan
Method	POST
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setWirelessLan
Example Request (json)	<pre> { "ssid":"AP1-2.4GHz", "type":"wpa2-psk(aes)", "key":"abadcafe", "dhcp":false, "ip_address":"192.168.0.11", "subnet_mask":"255.255.255.0", "default_gateway":"192.168.0.1", "dns_server_1":"192.168.0.1", "dns_server_2":"0.0.0.0" } </pre>

	}
--	---

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.6. setWirelessDirect

For setting Wireless Network (Wireless Direct). Network connection is switched to wireless (Wireless Direct) by using this API. If no parameter is specified, current parameter is used. If set parameter is incomplete, it is possible not to provide network availability.

Request Parameters

name	type	required	description
type	string	no	Specifies wireless encryption protocol Values: "none" / "wpa2-psk(aes)"
key	string	no	Specifies encryption key (printable ASCII (0x20-0x7E) within 64 characters). This value is invalid when type "none" is specified.

URI	<BaseURL>/v1/system/setWirelessDirect
Method	POST
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setWirelessDirect
Example Request (json)	{ "type":"wpa2-psk(aes)", "key":"abadcafe" }

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.7. setIpSettings

For setting IP. This API only set IP as maintain same network connection status (Wired/Wireless Lan/Wireless Direct/Extend). If no parameter is specified, current parameter is used. If set parameter is incomplete, it is possible not to provide network availability.

Request Parameters

name	type	required	description
dhcp	string	no	Specifies DHCP setting
ip_address	string	no	Specifies IP Address
subnet_mask	string	no	Specifies Subnet Mask
default_gateway	string	no	Specifies Default Gateway
dns_server_1	string	no	Specifies DNS Server 1
dns_server_2	string	no	Specifies DNS Server 2

URI	<BaseURL>/v1/system/setIpSettings		
Method	POST		
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setIpSettings		
Example Request (json)	<pre>{ "dhcp":false, "ip_address":"192.168.0.11", "subnet_mask":"255.255.255.0", "default_gateway":"192.168.0.1", "dns_server_1":"192.168.0.1", "dns_server_2":"0.0.0.0" }</pre>		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre>{ "response_code":0 }</pre>		
------------------	------------------------------------	--	--

4.8. setNetworkName

For setting Network Name (Friendly Name).

Request Parameters

name	type	required	description
name	string	yes	Specifies Network Name. Specifiable up to 32 characters.

URI	<BaseURL>/v1/system/setNetworkName		
Method	POST		
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setNetworkName		
Example Request (json)	<pre>{ "name":"Yamaha AVR" }</pre>		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details
Example Response	<pre>{ "response_code":0 }</pre>	

4.9. setAirPlayPin

For setting AirPlay PIN. This is valid only when “airplay” exists in “func_list” found in /system/getFeatures.

Request Parameters

name	type	required	description
pin	string	yes	Specifies AirPlay PIN (printable ASCII (0x20-0x7E) within 63 characters).

URI	<BaseURL>/v1/system/setAirPlayPin		
Method	POST		
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setAirPlayPin		
Example Request (json)	<pre>{ "pin": "abc123" }</pre>		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	<pre>{ "response_code":0 }</pre>	
------------------	------------------------------------	--

4.10. getMacAddressFilter

For retrieving setup of MAC Address Filter.

Request Parameters

URI	<BaseURL>/v1/system/getMacAddressFilter		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/getMacAddressFilter		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details
filter	boolean	Returns whether or not Filter is valid

address_1	string	Returns MAC Address [1] (12 digit ASCII)
...	...	...
address_10	string	Returns MAC Address [10] (12 digit ASCII)

Example Response	<pre>{ "response_code":0, "filter":true, "address_1":"00A0DE1BFFFA", "address_2":"00A0DE1BFFFB", "address_3":"00A0DE1BFFFC", "address_4":"00A0DE1BFFFD", "address_5":"00A0DE1BFFFE", "address_6":"000000000000", "address_7":"000000000000", "address_8":"000000000000", "address_9":"000000000000", "address_10":"00A0DE1BFFFF" }</pre>
------------------	--

4.11. setMacAddressFilter

For setting MAC Address Filter.

Request Parameters

name	type	required	description
filter	boolean	no	Specifies valid/invalid setup of Filter
address_1	string	no	Specifies MAC Address [1] (12 digit ASCII)
...	...	...	...
address_10	string	no	Specifies MAC Address [10] (12 digit ASCII)

URI	<BaseURL>/v1/system/setMacAddressFilter
Method	POST
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setMacAddressFilter
Example Request (json)	<pre>{ "filter":true, "address_1":"00A0DE1BFFFA", "address_2":"00A0DE1BFFFB", "address_3":"00A0DE1BFFFC", "address_10":"00A0DE1BFFFF" }</pre>

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details

Example Response	<pre>{ "response_code":0 }</pre>
------------------	------------------------------------

	}
--	---

4.12. getNetworkStandby

For retrieving setup of Network Standby.

Request Parameters

URI	<BaseURL>/v1/system/getNetworkStandby
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getNetworkStandby

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details
network_standby	string	Returns setup of Network Standby. "auto" is available only when "network_standby_auto" exists in system - func_list under /system/getFeatures Values: "off" / "on" / "auto"

Example Response	{ "response_code":0, "network_standby":"auto" }
------------------	--

4.13. setNetworkStandby

For setting Network Standby.

Request Parameters

name	type	required	description
standby	string	yes	Specifies setup of Network Standby. "auto" is available only when "network_standby_auto" exists in system - func_list under /system/getFeatures Values: "off" / "on" / "auto"

URI	<BaseURL>/v1/system/setNetworkStandby?standby=<standby>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setNetworkStandby?standby=on

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{
------------------	---

	<pre>"response_code":0 }</pre>
--	--------------------------------

4.14. getBluetoothInfo

For retrieving setup/information of Bluetooth. Parameters are readable only when corresponding functions are available in “func_list” of /system/getFeatures. “bluetooth_device” parameter is contained in “bluetooth_tx_setting”.

Request Parameters

URI	<BaseURL>/v1/system/getBluetoothInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getBluetoothInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details
bluetooth_standby	boolean	Returns setup of Bluetooth Standby
bluetooth_tx_setting	boolean	Returns Bluetooth transmission setting
bluetooth_device	object	Returns information of Bluetooth (Sink) device
- connected	boolean	Returns status of connection to device
- name	string	Returns device name (max size is 128 bytes).
- type	string	Returns type of device. Values: "loudspeaker" / "headphone" / "handsfree_device" / "unknown"
- address	string	Returns BT address (12-digit hex).
- scms_support	boolean	Returns status of SCMS compatibility of device

Example Response	<pre>{ "response_code":0, "bluetooth_standby":true, "bluetooth_tx_setting":true, "bluetooth_device":{ "connected":true, "name":"Yamaha Headphone", "type":"headphone", "address":"C259197BD6F5" "scms_support":true } }</pre>
------------------	---

4.15. setBluetoothStandby

For setting Bluetooth Standby.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies Bluetooth Standby setting

URI	<BaseURL>/v1/system/setBluetoothStandby?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setBluetoothStandby?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.16. setBluetoothTxSetting

For setting Bluetooth transmission.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies Bluetooth transmission setting

URI	<BaseURL>/v1/system/setBluetoothTxSetting?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setBluetoothTxSetting?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.17. getBluetoothDeviceList

For retrieving Bluetooth (Sink) device list. This API is available only when "bluetooth_tx_setting" is true under /system/getBluetoothInfo.

This device list information is in the cache. If update device list information, execute /system/updateBluetoothDeviceList.

Request Parameters

URI	<BaseURL>/v1/system/getBluetoothDeviceList
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getBluetoothDeviceList

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details
updating	boolean	Returns whether or not device list is being updated. Returns true while device list is being updated via /system/updateBluetoothDeviceList
device_list	array	Returns Bluetooth device list
-- name	string	Returns device name (max size is 128 bytes).
-- type	string	Returns type of device. Values: "loudspeaker" / "headphone" / "handsfree_device" / "unknown"
- address	string	Returns BT address (12-digit hex)
- deletable	boolean	Returns whether this device information can be removed from the list. If it's true, it can be deleted. If this information is not exists, treated as false.

Example Response	<pre>{ "response_code":0, "updating":false, "device_list":[{ "name":"Yamaha Speaker", "type":"loudspeaker", "address":"9AF2B8CE1580" }, ... { "name":"Yamaha Headphone", "type":"headphone", "address":"C259197BD6F5", "deletable":true }] }</pre>
------------------	--

4.18. updateBluetoothDeviceList

For updating Bluetooth (Sink) device list. This API is available only when “bluetooth_tx_setting” is true under /system/getFuncStatus.

Retrieve update status and list information after finish updating via /system/getBluetoothDeviceList.

Request Parameters

URI	<BaseURL>/v1/system/updateBluetoothDeviceList
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/updateBluetoothDeviceList

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error

		Refer to " Response Code List " for details
Example Response	<pre>{ "response_code":0 }</pre>	

4.19. connectBluetoothDevice

For connecting Bluetooth (Sink) device. This API is available only when "bluetooth_tx_setting" is true under /system/getBluetoothInfo.

It is possible to take time to return this API response issued after connection status is fixed.

In models where the value of system - bluetooth - update_cancelable of /system/getFeatures is true, connection processing can be performed even while updating the device list. (Device list update will be canceled)

If false or no parameter exists, connection processing can not be performed while updating the device list.

Request Parameters

name	type	required	description
address	string	yes	Specifies Bluetooth address

URI	<BaseURL>/v1/system/connectBluetoothDevice?address=<address>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/connectBluetoothDevice?address=C259197BD6F5		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre>{ "response_code":0 }</pre>	
------------------	------------------------------------	--

4.20. disconnectBluetoothDevice

For disconnecting Bluetooth (Sink) device. This API is available only when "bluetooth_tx_setting" is true under /system/getBluetoothInfo.

This API response is issued immediately after disconnect request is accepted.

Request Parameters

URI	<BaseURL>/v1/system/disconnectBluetoothDevice		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/disconnectBluetoothDevice		

Response Parameters

name	type	description
------	------	-------------

response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details
---------------	---------	--

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.21. deleteBluetoothDeviceHistory

Deletes the specified Bluetooth device information from the connection history stored in the device.

Only devices with the value of device_list - deletable in getBluetoothDeviceList is true can be deleted.

Request Parameters

name	type	required	description
address	string	yes	Specifies Bluetooth address

Request Parameters

URI	<BaseURL>/v1/system/deleteBluetoothDeviceHistory?address=<address>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/deleteBluetoothDeviceHistory?address=C259197BD6F5

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.22. getFuncStatus

For retrieving setup/information of overall system function. Parameters are readable only when corresponding functions are available in “func_list” of /system/getFeatures.

Request Parameters

URI	<BaseURL>/v1/system/getFuncStatus
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getFuncStatus

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details
auto_power_standby	boolean	Returns whether or not Auto Power Standby is enabled

ir_sensor	boolean	Returns whether or not IR sensor is available
speaker_a	boolean	Returns whether or not Speaker A is available
speaker_b	boolean	Returns whether or not Speaker B is available
headphone	boolean	Returns whether or not headphone is connected
dimmer	integer	Returns Dimmer setting number. Returns -1 if it's Auto Dimmer, and 0 or higher if its' manual setup
zone_b_volume_sync	boolean	Returns whether Zone B volume sync Zone A volume or not.
hdmi_out_1	boolean	Returns output setting of HDMI OUT 1 terminal
hdmi_out_2	boolean	Returns output setting of HDMI OUT 2 terminal
hdmi_out_3	boolean	Returns output setting of HDMI OUT 3 terminal.
auto_play	boolean	Returns setting of Auto Play
speaker_pattern	integer	Returns speaker setting pattern number Values: 1 - Speaker setting pattern number that can be acquired with /system/getFeatures
turn_table_rotation_time	integer	Returns turn table total rotation hours Values : 0 - 65535
party_mode	boolean	Returns status of Party Mode
video_preset	integer	Returns number of video preset Values: 1 - Video preset number that can be acquired with /system/getFeatures. If video_preset_disable is true and the feature is disabled, the value will be 1 (minimum value).
video_preset_disable	boolean	Returns whether or not Video Preset feature is disabled. Values: true (Video Preset function disabled), false (Video Preset function enabled). If this parameter does not exist, It is treated as the Video Preset function is enabled.
hdmi_standby_through	string	Returns setting of hdmi standby through Values: Hdmi standby through settings can be acquired with /system/getFeatures

Example Response	<pre>{ "response_code":0, "auto_power_standby":false, "ir_sensor":false, "speaker_a":true, "speaker_b":false, "headphone":false, "dimmer":3, "zone_b_volume_sync":false "hdmi_out_1":true, "hdmi_out_2":false "hdmi_out_3":false, "auto_play":"auto", "speaker_pattern":1, "turn_table_rotation_time":99, "party_mode":false, "video_preset":6, "video_preset_disable":false, "hdmi_standby_through":"auto" }</pre>
------------------	---

4.23. setAutoPowerStandby

For setting Auto Power Standby status. Actual operations/reactions of enabling Auto Power Standby depend on each Device.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies Auto Power Standby status

URI	<BaseURL>/v1/system/setAutoPowerStandby?enable=<enable>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/setAutoPowerStandby?enable=true		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }	
------------------	-----------------------------	--

4.24. setIrSensor

For setting remote control IR sensor.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies setting of IR sensor

URI	<BaseURL>/v1/system/setIrSensor?enable=<enable>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/setIrSensor?enable=true		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }	
------------------	-----------------------------	--

4.25. setSpeakerA

For setting Speaker A status.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies Speaker A status

URI	<BaseURL>/v1/system/setSpeakerA?enable=<enable>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/setSpeakerA?enable=true		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.26. setSpeakerB

For setting Speaker B status.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies Speaker B status

URI	<BaseURL>/v1/system/setSpeakerB?enable=<enable>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/setSpeakerB?enable=true		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.27. setDimmer

For setting FL/LED Dimmer.

Request Parameters

name	type	required	description
value	integer	yes	Setting Dimmer. Specifies -1 in case of auto setting. Specifies 0 or more than 0 in case of manual setting. Auto setting is available only when -1 is exists in vale range under /system/getFeatures.

			Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
--	--	--	---

URI	<BaseURL>/v1/system/setDimmer?value=<value>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setDimmer?value=3

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.28. setZoneBVolumeSync

For setting Zone B volume sync.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies whether or not Zone B volume sync Zone A volume.

URI	<BaseURL>/v1/system/setZoneBVolumeSync?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setZoneBVolumeSync?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.29. setHdmiOut1

For setting HDMI OUT 1 terminal output status.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies HDMI OUT 1 terminal output status

URI	<BaseURL>/v1/system/setHdmiOut1?enable=<enable>
-----	---

Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setHdmiOut1?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.30.setHdmiOut2

For setting HDMI OUT 2 terminal output status.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies HDMI OUT 2 terminal output status

URI	<BaseURL>/v1/system/setHdmiOut2?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setHdmiOut2?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.31.setHdmiOut3

For setting HDMI OUT 3 terminal output status.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies HDMI OUT 3 terminal output status

URI	<BaseURL>/v1/system/setHdmiOut3?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setHdmiOut3?enable=true

Response Parameters

name	type	description
------	------	-------------

response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
---------------	---------	--

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.32. getNameText

For retrieving text information of Zone, Input, Sound program. If they can be renamed, can retrieve text information renamed.

Request Parameters

name	type	required	description
Id	string	no	Specifies ID. If no ID is specified, retrieve all information of Zone, Input, Sound program. Refer to " ALL ID List " for details

URI	<BaseURL>/v1/system/getNameText?id=<id>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getNameText?id=main

Response Parameters (No ID is specified)

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
zone_list	array	Returns a list of Zone
- id	string	Returns Zone ID
- text	string	Returns text information related to Zone ID
input_list	array	Returns a list of Input
- id	string	Returns Input ID
- text	string	Returns text information related to Input ID
sound_program_list	array	Returns a list of Sound program
- id	string	Returns Sound Program ID
- text	string	Returns text information related to Sound Program ID

Example Response	{ "response_code":0, "zone_list":[{ "id":"main", "text":"Living Room" }, { "id":"zone2", "text":"Kitchen" }], }
------------------	---

	<pre> "input_list":[{ "id":"hdmi1", "text":"BD Player" }, { "id":"hdmi2", "text":"HDMI2" }, ... { "id":"spotify", "text":"Spotify" }], "sound_program_list":[{ "id":"munich", "text":"Hall_in_Munich" }, { "id":"vienna", "text":"Hall_in_Vienna" }, ... { "id":"straight", "text":"Straight" }] } </pre>
--	---

Response Parameters (ID is specified)

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details
id	string	Returns ID
text	string	Returns text information related to ID

Example Response	<pre> { "response_code":0, "id":"hdmi1", "text":"BD Player" } </pre>
------------------	--

4.33. setNameText

For setting text information related to each ID of Zone, Input.

Request Parameters

name	type	required	description
id	string	yes	Specifies ID. Input ID can be specified only when "rename_enable" is true under /system/getFeatures. Sound Program ID can not be specified. Note: If "main" is specified, Network Name is overwritten with same text information to be acceptable both MusicCast CONTROLLER (Yamaha) and Spotify App. If Network Name is changed, "main" text information is not changed.
text	string	yes	Specifies text information (UTF-8 within 64 bytes). If "" (empty text) is specified, specifies default text information.

URI	<BaseURL>/v1/system/setNameText		
Method	POST		
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setNameText		
Example Request (json)	{ "id":"main", "text":"Living Room" }		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.34. getNameText_AVR

Acquires text information (text information of the main unit (GUI / OSD / FL etc.)) corresponding to the zone ID.

Available only when "name_text_avr" exists in system_func_list of /system/getFeatures.

Dedicated to the Zone of AVR.

Request Parameters

name	type	required	Description
id	string	no	Specifies Zone ID. If no zone ID is specified, information on all zones can be collected at one time. Values : "main" / "zone2" / "zone3" / "zone4"

URI	<BaseURL>/v1/system/getNameText_AVR?id=<id>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/getNameText_AVR?id=main		

Response Parameters (No ID is specified)

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
zone_list	array	Returns Zone list
- id	string	Returns Zone ID
- text	string	Returns text information related to Zone ID.(Text information on the main unit (GUI / OSD / FL etc))

Example Response

```
{
  "response_code":0,
  "zone_list":[
    {
      "id":"main",
      "text":"Living"
    },
    {
      "id":"zone2",
      "text":"Kitchen"
    },
    {
      "id":"zone3",
      "text":"Bed Room1"
    },
    {
      "id":"zone4",
      "text":"Bed Room2"
    }
  ]
}
```

Response Parameters (ID is specified)

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
id	string	Returns Zone ID
text	string	Returns text information related to Zone ID.(Text information on the main unit (GUI / OSD / FL etc))

Example Response

```
{
  "response_code":0,
  "id":"main",
  "text":"Room1"
}
```

4.35.setNameText_AVR

Set text information (text information of the main unit (GUI / OSD / FL etc.)) corresponding to the zone ID.

Available only when "name_text_avr" exists in system_func_list of /system/getFeatures.
Dedicated to the Zone of AVR.

Request Parameters

name	type	required	Description
id	string	yes	Specifies Zone ID. Values : "main" / "zone2" / "zone3" / "zone4"
text	string	yes	Specify text information (text information of the main unit (GUI / OSD / FL etc)). Due to the limitations of the main unit, it is an error to enter anything other than Latin-1. Values: 1 to 9 characters included in Latin-1

URI	<BaseURL>/v1/system/setNameText_AVR
Method	POST
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setNameText_AVR
Example Request (json)	{ "id":"main", "text":"Living" }

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
-------------------------	-----------------------------

4.36. getLocationInfo

For retrieving Location information.

Request Parameters

URI	<BaseURL>/v1/system/getLocationInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getLocationInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
id	string	Returns Location ID in 32-digit hex
name	string	Returns Location Name
zone_list	object	Returns Location setup status of each Zone. Retrievable only when each Zone is valid

- main	boolean	Returns whether or not Main Zone Location setup is valid
- zone2	boolean	Returns whether or not Zone2 Location setup is valid
- zone3	boolean	Returns whether or not Zone3 Location setup is valid
- zone4	boolean	Returns whether or not Zone4 Location setup is valid
stereo_pair_status	string	Returns Stereo Pair setting status Values : "none" / "master_left" / "master_right" / "slave_left" / "slave_right"

Example Response	<pre>{ "response_code":0, "id":"9A237BF5AB80ED3C7251DFF49825CA42", "name":"Home", "zone_list":{ "main":true, "zone2":true, "zone3":false }, "stereo_pair_status":"none" }</pre>
------------------	---

4.37. getStereoPairInfo

For retrieving Stereo Pair information.

Request Parameters

URI	<BaseURL>/v1/system/getStereoPairInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getStereoPairInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
status	string	Returns Stereo Pair status Values : "none" / "master_left" / "master_right" / "slave_left" / "slave_right"
pair_info	object	Returns the other side device (Stereo Pair) information
- alive	boolean	Returns whether or not the other side device is alive
- ip_address	string	Returns IP Address
- mac_address	string	Returns MAC Address (Wireless Direct 12 digit ASCII)

Example Response	<pre>{ "response_code":0, "status":"master_left", "pair_info":{ "alive":true, "ip_address":"192.168.0.10", "mac_address":"00A0DE012345" } }</pre>
------------------	---

	}
--	---

4.38. sendIrCode

For sending specific remote IR code. A Device is operated same as remote IR code reception. But continuous IR code cannot be used in this command. Refer to each Device's IR code list for details

Request Parameters

name	type	required	description
code	string	yes	Specifies IR code in 8-digit hex

URI	<BaseURL>/v1/system/sendIrCode?code=<code>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/sendIrCode?code=7F016C13

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.39. getRemoteInfo

For retrieving remote monitor information. Available only when " remote_info " exists in system - func_list under /system/getFeatures.

URI	<BaseURL>/v1/system/getRemoteInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getRemoteInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
model_name	string	Returns model name. Same value as /system/getDeviceInfo - model_name Ex : "RX-V479" / "WXC-50" / "UNKNOWN" etc.
destination	string	Returns Device's destination region code. Same value as /system/getDeviceInfo - destination Values: "J" / "U" / "C" / "R" / "B" / "G" / "BG" / "T" / "A" / "L" / "K" / "E" / "V" / "F" / "P" / "S" / "H"
manufacture	string	"Yamaha Corporation"
system_version	floating point	Returns System Version . Same as /system/getDeviceInfo - system_version
api_version	floating point	Returns API Version.

		Same as /system/getDeviceInfo - api_version
netmodule_version	string	Returns Network Module version. Same as /system/getDeviceInfo - netmodule_version
netmodule_checksum	string	Return check-sum of Network Module in hex. Same value as /system/getDeviceInfo - netmodule_checksum
serial_number	string	Returns Serial Number(16 alphanumeric characters)
ip_address	string	Returns IP Address. Same value as /system/getNetworkStatus - ip_address
mac_address	object	Returns MAC Address (12 digit ASCII). Same value as /system/getNetworkStatus - mac_address
- wired_lan	string	Returns wired MAC Address
- wireless_lan	string	Returns wireless (Wi-Fi) MAC Address
- wireless_direct	string	Returns wireless (Wireless Direct) MAC Address
network_standby	string	Returns setup of Network Standby. Same value as /system/getNetworkStandby - network_standby Values : "off" / "on" / "auto"
main	object	Returns Main Zone related information
- power	string	Returns power status. Same value as /main/getStatus - power Value : "on" / "standby"
- input	string	Returns selected Input ID. Same value as /main/getStatus - input Values: Input IDs gotten via /system/getFeatures
zone2	object	Returns information related to Zone2. This data format is same as main zone's one above.
zone3	object	Returns information related to Zone3. This data format is same as main zone's one above.
zone4	object	Returns information related to Zone4. This data format is same as main zone's one above.
protection_history	string array	Returns protection history (max size is 8 and top is the latest one)

Example Response	<pre>{ "response_code":0, "model_name":"RX-V483", "destination":"U", "manufacture":"Yamaha Corporation", "system_version":1.10, "api_version":2.00, "netmodule_version":"0200 ", "netmodule_checksum":"878059DD", "serial_number":"123456789abcdef", "ip_address":"192.168.0.10", "mac_address":{ "wired_lan":"00A0DE012345", "wireless_lan":"78A501012345", "wireless_direct":"78A501012346" }, "network_standby":"auto", "main":{</pre>
------------------	---

	<pre> "power":"on", "input":"siriusxm" }, "zone2":{ "power":"on", "input":"cd" }, "protection_history":["DC", "PS2", ... "..."] } </pre>
--	--

4.40. requestNetworkReboot

For requesting Network module reboot. Available only when " network_reboot " exists in system - func_list under /system/getFeatures.

Request Parameters

URI	<BaseURL>/v1/system/requestNetworkReboot
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/requestNetworkReboot

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre> { "response_code":0 } </pre>
------------------	--------------------------------------

4.41. requestSystemReboot

For requesting System reboot. Available only when " system_reboot " exists in system - func_list under /system/getFeatures.

Request Parameters

URI	<BaseURL>/v1/system/requestSystemReboot
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/requestSystemReboot

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.42. getAdvancedFeatures

Get additional information about the functions installed in the device.

Request Parameters

URI	<BaseURL>/v1/system/getAdvancedFeatures
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/getAdvancedFeatures

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
web_settings_list	string array	Returns the items displayed on the Web Settings screen. Items not to be displayed are not returned. Values : "network_name" / "network_settings" / "airplay" / "firmware_update" / "standby" / "mac_filter" / "licenses" / "backup_restore" / "advanced_settings"
advanced_settings_list	array	If web_settings_list has advanced_settings, it returns information about that display. If there is no, this parameter does not exist. Returns the display items of the YXC version Advanced Settings screen.
- id	string	Returns items displayed on the Advanced Settings screen. Values : "system_reboot" / "auto_play" / "ir_sensor"
- link_list	string array	Returns information on whether to display on the Advanced Settings screen. "for_websettings" is displayed when linked from Web Settings "for_musiccast_app" is displayed when it is linked from MusicCast App Values : "for_websettings" / "for_musiccast_app"

Example Response	{ "response_code":0, "web_settings_list":["network_name", "network_settings", "airplay", "firmware_update", "standby", "mac_filter", "licenses", "backup_restore", "advanced_settings"], }
------------------	---

	<pre> "advanced_settings_list":[{ "id":"system_reboot", "link_list":["for_websettings"] }, { "id":"auto_play", "link_list":["for_websettings"] }, { "id":"ir_sensor", "link_list":["for_websettings", "for_musiccast_app"] }] </pre>
--	--

4.43. setAutoPlay

For setting Auto Play of device. Available only when " auto play function " exists in system - func_list under /system/getFeatures.

Request Parameters

name	type	Required	description
enable	boolean	Yes	Specifies Auto Play setting Values : true (enable) , false (disable)

URI	<BaseURL>/v1/system/setAutoPlay?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setAutoPlay?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre> { "response_code":0 } </pre>
------------------	--------------------------------------

4.44. setSpeakerPattern

For setting the speaker setting pattern. Available only when " speaker_pattern function " exists

in system - func_list under /system/getFeatures.

Request Parameters

name	type	required	description
num	integer	yes	Specifies Speaker pattern number. Values : 1～speaker_pattern number from /system/getFeatures

URI	<BaseURL>/v1/system/setSpeakerPattern?num=<num>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/setSpeakerPattern?num=1		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }	
------------------	-----------------------------	--

4.45. clearTurnTableRotationTime

Clear the total turning hours of the turntable.

It can be executed only when the turn_table_rotation_time function is enabled in system - func_list of /system/getFeatures.

Request Parameters

URI	<BaseURL>/v1/system/clearTurnTableRotationTime		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/system/clearTurnTableRotationTime		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }	
------------------	-----------------------------	--

4.46. setPartyMode

For setting Party Mode. Available only when " party_mode " exists in system - func_list under /system/getFeatures.

Request Parameters

Name	type	required	description
Enable	boolean	yes	Specifies Party Mode setting.

			Values : true (Enable), false (disable)
--	--	--	---

URI	<BaseURL>/v1/system/setPartyMode?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setPartyMode?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.47. setPartyVolume

Set the volume of zones participating in party mode at one time.

Available only when "party_volume" exists in system-func_list of /system/getFeatures.

Request Parameters

name	type	required	description
value	string	no	Specifies Volume. Values : "up" / "down"

URI	<BaseURL>/v1/system/setPartyVolume?value=<value>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setPartyVolume?value=up

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.48. setPartyMute

Set the mute of zones participating in party mode at one time.

Available only when "party_mute" exists in system-func_list of /system/getFeatures.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies mute status.

URI	<BaseURL>/v1/system/setPartyMute?enable=<enable>
-----	--

Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setPartyMute?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.49. setVideoPreset

Set the video preset number.

Available only when "video_preset" exists in system_func_list of /system/getFeatures.

Request Parameters

name	type	required	description
num	integer	yes	Specify video preset number. Value: 1 to upper limit defined in /system/getFeatures (video_preset_num)

URI	<BaseURL>/v1/system/setVideoPreset?num=<num>
Method	GET
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/system/setVideoPreset?num=1

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details.

Example Response	{ "response_code":0 }
------------------	-----------------------------

4.50. setHdmiStandbyThrough

Set HDMI Standby Through.

Available only when "hdmi_standby_through" exists in system_func_list of /system/getFeatures.

Request Parameters

name	type	required	description
value	string	yes	Values: Hdmi standby through settings can be acquired with /system/getFeatures

URI	<BaseURL>/v1/system/setHdmiStandbyThrough?value=<value>
-----	---

Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/system/setHdmiStandbyThrough?value=on

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details.

Example Response	{ "response_code":0 }
------------------	-----------------------------

5. Zone

5.1. getStatus

For retrieving basic information of each Zone like power, volume, input and so on.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone Values: "main" / "zone2" / "zone3" / "zone4"

URI	<BaseURL>/v1/<zone>/getStatus
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/getStatus

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details
power	string	Returns power status Values: "on" / "standby"
sleep	integer	Returns Sleep Timer setup value (unit in minutes) Values: 0 / 30 / 60 / 90 / 120
volume	integer	Returns volume value Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
mute	boolean	Returns mute status
max_volume	integer	Returns Max Volume setup Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
input	string	Returns selected Input ID Values: Input IDs gotten via /system/getFeatures
input_text	string	Returns text information selected Input ID
distribution_enable	boolean	Returns whether or not current Input is distributable status
sound_program	string	Returns selected Sound Program ID Values: Sound Program IDs gotten via /system/getFeatures
surr_decoder_type	string	Returns selected Surround Decoder Type Values: Surround Decoder Type gotten via /system/getFeatures
surround_ai	boolean	Returns SURROUND: AI status
surround_3d	boolean	Returns 3D Surround status
direct	boolean	Returns Direct status
pure_direct	boolean	Returns Pure Direct status
enhancer	boolean	Returns Enhancer status
tone_control	object	Returns Tone Control status
- mode	string	Returns Mode type. Available on after API version 1.15 Values: what is gotten via /system/getFeatures
- bass	integer	Returns Bass status

		Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
- treble	integer	Returns Treble status Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
equalizer	object	Returns Equalizer status
- mode	string	Returns Mode type. Available on after API version 1.15 Values: what is gotten via /system/getFeatures
- low	integer	Returns Low status Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
- mid	integer	Returns Mid status Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
- high	integer	Returns High status Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
balance	integer	Returns Speaker L/R Balance status. Negative values are for left side, positive values are for right side balance Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
dialogue_level	integer	Returns Dialogue Level status Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures
dialogue_lift	integer	Returns Dialogue Lift status Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
clear_voice	boolean	Returns Clear Voice status
subwoofer_volume	integer	Returns Subwoofer Volume status Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures
bass_extension	boolean	Returns Bass Extension status
link_control	string	Returns Link Control setup status Values: what is gotten via /system/getFeatures
link_audio_delay	string	Returns Audio Delay setup status Values: what is gotten via /system/getFeatures
link_audio_quality	string	Returns audio quality setting of distribution Values: what is gotten via /system/getFeatures
disable_flags	integer	Returns flags of functions that are not operable/available currently. Disable status is expressed as OR of bit field shown below, i.e. functions that listed in zone - func_list of /system/getFeatures, and also that have no flags of below are only valid/available/operable b[0] Volume b[1] Mute b[2] Link Audio Delay b[3] Sound Program
contents_display	boolean	Returns display setting of screen Not displayed if contents_display does not exist in the zone - func_list of /system/getFeatures.

actual_volume	object	Returns the display volume setting. Not displayed when there is no volume.
- mode	string	Returns current Actual Volume Mode. Values: actual_volume_mode_list gotten via /system/getFeatures
- value	floating point	Returns display volume value
- unit	string	Returns display volume unit. If there is no unit, returns an empty string
audio_select	string	Returns current Audio Select. Values: audio_select_list gotten via /system/getFeatures
party_enable	boolean	Returns party status of zone
extra_bass	boolean	Returns setting of extra bass
adaptive_drc	boolean	Returns setting of adaptive drc
dts_dialogue_control	integer	Returns setting of dts dialogue control Values: value calculated from the minimum value, maximum value, and step value gotten via /system/getFeatures
mono	boolean	Returns setting of mono
adaptive_dsp_level	boolean	Returns setting of adaptive dsp level
bluetooth_tx	object	Returns information about Bluetooth transmission. It can be obtained when bluetooth_device-connected of / system / getBluetoothInfo is true.
- audio_out	boolean	Returns audio output state to Bluetooth.
- restriction	string	Returns the reason why audio can not be output to Bluetooth. It can be obtained when audio_out is false. Values: "dist_prohibited" / "no_scms_support"

Example Response	<pre>{ "response_code":0, "power":"on", "sleep":0, "volume":30, "mute":false, "max_volume":100, "input":"pandora", "input_text":"Pandora", "distribution_enable":true, "sound_program":"chamber", "surr_decoder_type":"auto", "surround_ai":false, "surround_3d":false, "direct":false, "pure_direct":false, "enhancer":true, "tone_control":{ "bass":0, "treble":2 }, "dialogue_level":0, "dialogue_lift":1, "disable_flags":0 }</pre>
------------------	---

	<pre> "contents_display":true, "actual_volume":{ "mode":"db", "value":-23.5, "unit":"dB" }, "audio_select":"auto", "party_enable":false, "extra_bass":true, "adaptive_drc":true, "dts_dialogue_control":0, "mono":false, "adaptive_dsp_level":true, "bluetooth_tx":{ "audio_out":true "restriction": "dist_prohibited" } } </pre>
--	---

5.2. getSoundProgramList

For retrieving a list of Sound Program available in each Zone. It is possible for the list contents to be dynamically changed.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target Zone Values: "main" / "zone2" / "zone3" / "zone4"

URI	<BaseURL>/v1/<zone>/getSoundProgramList
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/getSoundProgramList

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details
sound_program_list	string array	Returns a list of selectable sound program

Example Response	<pre> { "response_code":0, "sound_program_list":["munich", "vienna", ... "straight"] } </pre>
------------------	---

5.3. setPower

For setting power status of each Zone.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
power	string	yes	Specifies power status Values: "on" / "standby" / "toggle"

URI	<BaseURL>/v1/<zone>/setPower?power=<power>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/zone3/setPower?power=on

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.4. setSleep

For setting Sleep Timer for each Zone.

With Zone B enabled Devices, only Main Zone can be used.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
sleep	integer	yes	Specifies Sleep Time (unit in minutes) Values: 0 / 30 / 60 / 90 / 120

URI	<BaseURL>/v1/<zone>/setSleep?sleep=<sleep>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setSleep?sleep=120

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

	}
--	---

5.5. setVolume

For setting volume in each Zone. Values of specifying range and steps are different. There are some Devices that cannot allow this value to be go up to Device's maximum volume.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
volume	integer	yes	Specifies volume value Value Range: calculated by minimum/maximum/step values gotten via /system/getFeatures (Available on and after API Version 1.17) "up" / "down"
step	integer	no	Specifies volume step value if the volume is "up" or "down". If nothing specified, minimum step value is used implicitly. (Available on and after API Version 1.17) Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setVolume?volume=<volume>&step=<step>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setVolume?volume=50

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.6. setMute

For setting mute status in each Zone.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifying mute status

URI	<BaseURL>/v1/<zone>/setMute?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setMute?enable=false

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.7. setInput

For selecting each Zone input.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone Values: "main" / "zone2" / "zone3" / "zone4"
input	string	yes	Specifies Input ID Values: Input IDs gotten via /system/getFeatures
mode	string	no	Specifies select mode. If no parameter is specified, actions of input change depend on a Device's specification. Value: "autoplay_disabled" (Restricts Auto Play of Net/USB related Inputs). Available on and after API Version 1.12

URI	<BaseURL>/v1/<zone>/setInput?input=<input>&mode=<mode>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setInput?input=airplay&mode=autoplay_disabled

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.8. setSoundProgram

For selecting Sound Programs.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
program	string	yes	Specifies Sound Program ID Values: Sound Program IDs gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setSoundProgram?program=<program>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setSoundProgram?program=vienna

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.9. setSurroundAI

For setting SURROUND:AI.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. It is possible to specify only the Zone where the Direct function exists. Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies settings of SURROUND:AI If SURROUND:AI setting is enabled, Sound Program setting will be disabled.

URI	<BaseURL>/v1/<zone>/setSurroundAI?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setSurroundAI?enable=true

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.10. set3dSurround

For setting 3D Surround status.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies 3D Surround status

URI	<BaseURL>/v1/<zone>/set3dSurround?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/set3dSurround?enable=true

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.11. setDirect

For setting Direct status.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies Direct status

URI	<BaseURL>/v1/<zone>/setDirect?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setDirect?enable=true

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.12. setPureDirect

For setting Pure Direct status.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies Pure Direct status

URI	<BaseURL>/v1/<zone>/setPureDirect?enable=<enable>
-----	---

Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setPureDirect?enable=true

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.13. setEnhancer

For setting Enhancer status.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies Enhancer status

URI	<BaseURL>/v1/<zone>/setEnhancer?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setEnhancer?enable=true

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.14. setToneControl

For setting Tone Control in each Zone. Values of specifying range and steps are different.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
mode	string	no	Specifies Mode setting. If no parameter is specified, current Mode setting is not changed. Regardless of the Mode setting, bass/treble setting can be changed, but valid only when Mode setting is "manual".

			Values: Values gotten via /system/getFeatures
bass	integer	no	Specifies Bass value Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures
treble	integer	no	Specifies Treble value Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setToneControl?mode=<mode>&bass=<base>&treble=<treble>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/main/setToneControl?mode=manual&bass=5&treble=-2		

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.15. setEqualizer

For setting Equalizer in each Zone. Values of specifying range and steps are different.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
mode	string	no	Specifies Mode setting. If no parameter is specified, current Mode setting is not changed. Regardless of the Mode setting, low/mid/high setting can be changed, but valid only when Mode setting is "manual". Values: Values gotten via /system/getFeatures
low	integer	no	Specifies Low value Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures
mid	integer	no	Specifies Mid value Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures
high	integer	no	Specifies High value Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setEqualizer?mode=<mode>&low=<low>&mid=<mid>&high=<high>		
Method	GET		
Example	http://{host}/YamahaExtendedControl/v1/main/setEqualizer?mode=manual&low=3&mid=-2&high		

Request	h=5
---------	---------------------

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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5.16. setBalance

For setting L/R Balance in each Zone's speaker. Values of specifying range and steps are different.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
value	integer	yes	Specifies L/R Balance value. Negative values are for left side, positive values are for right side balance. Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setBalance?value=<value>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setBalance?value=5

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.17. setDialogueLevel

For setting Dialogue Level in each Zone. Values of specifying range and steps are different.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
value	integer	yes	Specifies Dialogue Level value Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setDialogueLevel?value=<value>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setDialogueLevel?value=2

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.18. setDialogueLift

For setting Dialogue Lift in each Zone. Values of specifying range and steps are different.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
value	integer	yes	Specifies Dialogue Lift value Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setDialogueLift?value=<value>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setDialogueLift?value=3

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.19. setClearVoice

For setting Clear Voice in each Zone.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies Clear Voice setting

URI	<BaseURL>/v1/<zone>/setClearVoice?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setClearVoice?enable=false

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.20. setSubwooferVolume

For setting Subwoofer Volume in each Zone.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
volume	integer	yes	Specifies volume value Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setSubwooferVolume?volume=<volume>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setSubwooferVolume?volume=3

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.21. setBassExtension

For setting Bass Extension in each Zone.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target zone. Available for zones with this function Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies Bass Extension setting

URI	<BaseURL>/v1/<zone>/setBassExtension?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setBassExtension?enable=false

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.22. getSignalInfo

For retrieving current playback signal information in each Zone.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone Values: "main" / "zone2" / "zone3" / "zone4"

URI	<BaseURL>/v1/<zone>/getSignalInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/getSignalInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
audio	object	Returns Audio related information
- error	integer	Returns error information Values: 0: No Error 1: Unknown Error 2: Check Cable (USB DAC) 3: Check PC (USB DAC) 4: Disconnected (USB DAC) 5: Check Signal : Over Fs 384kHz (USB DAC)
- format	string	Returns audio format. Returns "" (empty text) in inputs unsupported to retrieve audio format. Values: "" / "----" (Unknown) / "Analog" / "Digital" / "PCM" / "MPCM" / "WAV" / "FLAC" / "MP3" / "WMA" / "AAC" / "ALAC" / "AIFF" / "DSD" / "Vorbis" / "SBC"
- fs	string	Returns Fs. Returns "" (empty text) in inputs unsupported to retrieve Fs. Values: "" / "----" (Unknown) / "Analog" / "8 kHz" / "11.025 kHz" / "12 kHz" / "16 kHz" / "22.05 kHz" / "24 kHz" / "32 kHz" / "44.1 kHz" / "48 kHz" / "64 kHz" / "88.2 kHz" / "96 kHz" / "128 kHz" / "176.4 kHz" / "192 kHz" / "352.8 kHz"

	kHz" / "384 kHz" / "2.8 MHz" / "5.6 MHz" / "11.2 MHz"
Example Response	<pre>{ "response_code":0, "audio":{ "error":0, "format":"FLAC", "fs":"192 kHz" } }</pre>

5.23. prepareInputChange

Let a Device do necessary process before changing input in a specific zone. This is valid only when “prepare_input_change” exists in zone - func_list found in /system/getFeatures.

MusicCast CONTROLLER executes this API when an input icon is selected in a Room, right before sending various APIs (of retrieving list information etc.) regarding selecting input.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target zone Values: "main" / "zone2" / "zone3" / "zone4"
input	string	yes	Specifies Input ID Values: Input IDs gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/prepareInputChange?input=<input>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/prepareInputChange?input=usb

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	<pre>{ "response_code":0 }</pre>
------------------	------------------------------------

5.24. recallScene

Scene selection from the scene number.

It is valid only when "scene" exists in zone - func_list of /system/getFeatures.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target zone Values: "main" / "zone2" / "zone3" / "zone4"
num	integer	yes	Specifies scene number

			Values: Scene number gotten via /system/getFeatures
URI	<BaseURL>/v1/<zone>/recallScene?num=<num>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/main/recallScene?num=1		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0, }
------------------	------------------------------

5.25. setContentsDisplay

Set the screen display of each Zone's screen.

It is valid only when "contents_display" exists in zone - func_list of /system/getFeatures.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies display status of screen

URI	<BaseURL>/v1/<zone>/setContentsDisplay?enable=<enable>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/main/setContentsDisplay?enable=false		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.26. controlCursor

Operate the cursor keys on the remote control.

It is valid only when "cursor" exists in zone - func_list of /system/getFeatures.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone Values: "main" / "zone2" / "zone3" / "zone4"

cursor	string	yes	Specifies key to be executed Values: Cursor_list gotten via /system/getFeatures
--------	--------	-----	--

URI	<BaseURL>/v1/<zone>/controlCursor?cursor=<cursor>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/main/controlCursor?cursor=up		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0, }
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5.27. controlMenu

Operate the menu keys on the remote control.

It is valid only when " menu " exists in zone - func_list of /system/getFeatures.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone Values: "main" / "zone2" / "zone3" / "zone4"
menu	string	yes	Specifies key to be executed Values: Menu_list gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/controlMenu?menu=<menu>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/main/controlMenu?menu=top_menu		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0, }
------------------	------------------------------

5.28. setActualVolume

Set the volume of each Zone with the value to display.

It is valid only when " actual_volume " exists in zone - func_list of /system/getFeatures.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target Zone Values: "main" / "zone2" / "zone3" / "zone4"
mode	string	yes	Specifies mode Values: Actual_volume_mode_list gotten via /system/getFeatures
value	floating point	no	Specifies volume Values: Calculated from minimum value, maximum value, step value of volume gotten via /system/getFeatures
operation	string	no	(Valid in API Version 2.03 or later, and valid in value of netmodule_generation 2 or later gotten via /system/getFeatures) "up" / "down"
step	floating point	no	(Valid in API Version 2.03 or later, and valid in value of netmodule_generation 2 or later gotten via /system/getFeatures) Specifies step value of "up" / "down". If no value is specified, the minimum step value is treated as specified. Values: Calculated from minimum value, maximum value, step value of step gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setActualVolume?mode=<mode>&value=<value>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setActualVolume?mode=db&value=-23.5

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.29. setAudioSelect

Set the audio input selection.

It is valid only when "audio_select" exists in zone_func_list of /system/getFeatures.

In the value of audio_select_list obtained by /system/getFeatures, it is possible to specify something other than unavailable. If current audio_select (gotten with getStatus) is unavailable, it can not be set.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target Zone Values: "main" / "zone2" / "zone3" / "zone4"
type	string	yes	Specifies Audio Select Values: Audio_decoder_list gotten via /system/getFeatures (other than unavailable)

URI	<BaseURL>/v1/<zone>/setAudioSelect?type=<type>
-----	--

Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setAudioSelect?type=auto

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.30. setSurroundDecoderType

Set the Sound Program : Decoder Type to be used with Surround Decoder.

It is valid only when "surr_decoder_type" exists in zone - func_list of /system/getFeatures.

Request Parameters

name	type	required	Description
zone	string	yes	Specifies target Zone Values: "main" / "zone2" / "zone3" / "zone4"
type	string	yes	Specifies Surround Decoder Type Values: surr_decoder_type_list gotten via /system/getFeatures

URI	<BaseURL>/v1/<zone>/setAudioSelect?type=<type>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/main/setSurroundDecoderType?type=auto

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.31. setSceneInfo

Set the text information and tag information of the scene.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone. Available for zones with this function. Values: "main" / "zone2" / "zone3" / "zone4"
num	integer	yes	Specifies scene num. Values: 1 - zone's scene number gotten via /system/getFeatures
text	string	no	Specify tag information.

			Specify within UTF-8 128 bytes. If "" (empty string) is specified, default text information is stored.
tag	integer	no	Specifies tag information. Values: 0 - 255

URI	<BaseURL>/v1/<zone>/setSceneInfo		
Method	POST		
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/main/setSceneInfo		
Example Request (json)	<pre>{ "num":1, "text":"BD/DVD", "tag": 1, }</pre>		

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details

Example Response	<pre>{ "response_code":0 }</pre>
------------------	------------------------------------

5.32. getSceneInfo

Get the text and tag information of the scene.

If renaming is possible, it can obtain one that reflects the renaming settings.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone. Available for zones with this function. Values: "main" / "zone2" / "zone3" / "zone4"
num	integer	no	Specifies scene num. If it does not specify a scene number, it can get information on all scenes at once.

URI	<BaseURL>/v1/<zone>/getSceneInfo?num=<num>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/main/getSceneInfo?num=1		

Response Parameters (No scene number specified)

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details
scene_list	array	Returns scene list
- num	integer	Returns scene number
- text	string	Returns text information corresponding to the scene number

- tag	integer	Returns tag information
- default_tag	integer	Returns when the tag value is 0, the value to be used by the control person. If this parameter does not exist, treat it as being set to 0.

Example Response	<pre>{ "response_code":0, "scene_list":[{ "num":1, "text":"Movie Viewing", "tag": 1, "default_tag": 1 }, { "num":2, "text":"Radio Listening", "tag": 2, "default_tag": 2 }, { "num":3, "text":"Music Listening", "tag": 3, "default_tag": 3 }, { "num":4, "text":"NET Audio Listening", "tag": 4, "default_tag": 4 }, { "num":5, "text":"STB Viewing", "tag": 5, "default_tag": 5 }, { "num":6, "text":"Game Playing", "tag": 6, "default_tag": 6 }, { "num":7, "text":"Movie Viewing", "tag": 7, "default_tag": 7 }] }</pre>
------------------	---

	<pre> }, { "num":8, "text":"Media Server Listening", "tag": 8, "default_tag": 8 }] }</pre>
--	---

Response Parameters (scene number specified)

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details
num	integer	Returns scene number
text	string	Returns text information corresponding to the scene number
tag	integer	Returns tag information
default_tag	integer	Returns when the tag value is 0, the value to be used by the control person. If this parameter does not exist, treat it as being set to 0.

Example Response	<pre> { "response_code":0, "num":1, "text":"Movie Viewing", "tag": 1, "default_tag": 1 }</pre>
------------------	--

5.33. setExtraBass

Set Extra Bass.

Only the zone where "extra_bass" exists in zone-func_list of / system / getFeatures is valid.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone. Available for zones with this function. Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies Extra Bass settings. Values: true / false

URI	<BaseURL>/v1/<zone>/setExtraBass?enable=<enable>
Method	GET
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/main/setExtraBass?enable=true

Response Parameters

name	type	description
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response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details
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Example Response	{ "response_code":0 }
------------------	-----------------------------

5.34. setAdaptiveDrc

Set Adaptive DRC.

Only the zone where "adaptive_drc" exists in zone - func_list of /system/getFeatures is valid.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone. Available for zones with this function. Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies Adaptive DRC settings. Values: true / false

URI	<BaseURL>/v1/<zone>/setAdaptiveDrc?enable=<enable>
Method	GET
Example Request (URI)	http://<host>/YamahaExtendedControl/v1/main/setAdaptiveDrc?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.35. setDtsDialogueControl

Set DTS Dialogue Control.

Only the zone where "dts_dialogue_control" exists in zone - func_list of /system/getFeatures is valid.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone. Available for zones with this function. Values: "main" / "zone2" / "zone3" / "zone4"
num	integer	yes	Specifies DTS Dialogue Control settings. Values: 0 - Upper limit defined in /system/getFeatures

URI	<BaseURL>/v1/<zone>/setDtsDialogueControl?num=<num>
Method	GET

Example Request (URI)	http://{host}/YamahaExtendedControl/v1/main/setDtsDialogueControl?num=6
-----------------------	---

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.36. setMono

Set Mono.

Only the zone where "mono" exists in zone - func_list of /system/getFeatures is valid.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone. Available for zones with this function. Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies Mono settings. Values: true / false

URI	<BaseURL>/v1/<zone>/setMono?enable=<enable>
Method	GET
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/main/setMono?enable=true

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

5.37. setAdaptiveDspLevel

Set Adaptive DSP Level.

Only the zone where "adaptive_dsp_level" exists in zone - func_list of /system/getFeatures is valid.

Request Parameters

name	type	required	description
zone	string	yes	Specifies target Zone. Available for zones with this function.

			Values: "main" / "zone2" / "zone3" / "zone4"
enable	boolean	yes	Specifies Adaptive DSP Level settings. Values: true / false

URI	<BaseURL>/v1/<zone>/setAdaptiveDspLevel?enable=<enable>		
Method	GET		
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/main/setAdaptiveDspLevel?enable=true		

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to “ Response Code List ” for details

Example Response	{ "response_code":0 }		
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6. Tuner

APIs in regard to Tuner setting and getting information.

Target inputs: AM / FM / DAB

6.1. getPresetInfo

For retrieving Tuner preset information.

Request Parameters

name	type	required	description
band	string	yes	Specifying a band. Values depend on Preset Type gotten via /system/getFeatures. Values: "common" (common), "am" / "fm" / "dab" (separate)

URI	<BaseURL>/v1/tuner/getPresetInfo?band=<band>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/getPresetInfo?band=fm

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
preset_info	array	Returns preset information. A number of array elements is to be retrieved via /system/getFeature
- band	string	Returns band information. "unknown" when no presets Values: "am" / "fm" / "dab" / "unknown"
- number	integer	Returns; frequency (unit in kHz) (band = AM or FM) Station ID (band = DAB) 0 when there's no presets
- text	string	Returns text information of FM(RDS) or DAB
func_list	string array	Returns a list of valid functions for Preset. (Recall/Store functions are always valid without specifically listed here) Values: "clear" / "move"

Example Response	<pre>{ "response_code":0, "preset_info":[{ "band":"am", "number":530 "hd_program":0 "text":"" }, {</pre>
------------------	---

	<pre> "band": "fm", "number": 87500, "hd_program": 0 "text": "BBC1" }, ... { "band": "unknown", "number": 0, }], "func_list": ["clear", "move"] } </pre>
--	---

6.2. getPlayInfo

For retrieving playback information of Tuner.

Request Parameters

URI	<BaseURL>/v1/tuner/getPlayInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/getPlayInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details
band	string	Returns current Band Values: "am" / "fm" / "dab"
auto_scan	boolean	Returns Auto Scan (Up or Down) status
auto_preset	boolean	Returns Auto Preset execution status
am	object	Returns AM related information
- preset	integer	Returns current preset number. 0 when there's no presets Values: 0 (no presets), or one in the range gotten via /system/getFeatures
- freq	integer	Returns frequency (unit in kHz)
- tuned	boolean	Returns Tuned status
fm	object	Returns FM related information
- preset	integer	Returns current preset number. 0 when there's no presets Values: 0 (no presets), or one in the range gotten via /system/getFeatures
- freq	integer	Returns frequency (unit in kHz).
- tuned	boolean	Returns Tuned status
- audio_mode	string	Returns Audio Mode Values: "mono" / "stereo"
rds	object	Returns RDS information. Available only when RDS is valid
- program_type	string	Returns Program Type

- program_service	string	Returns Program Service
- radio_text_a	string	Returns Radio Text A
- radio_text_b	string	Returns Radio Text B
dab	object	Returns DAB related information. Available only when DAB is valid
- preset	integer	Returns current preset number. 0 when current station is not in presets. Values: 0 (no presets), or one in the range gotten via /system/getFeatures
- id	integer	Returns Station ID
- status	string	Returns DAB status. When it's in Tune Aid, valid parameters are "tune_aid" and "CH Label" only Values: "not_ready" / "initial_scan" / "tune_aid" / "ready"
- freq	integer	Returns DAB frequency (unit in kHz) Value Range: 174000 - 240000
- category	string	Returns Category Values: "primary" / "secondary"
- audio_mode	string	Returns Audio Mode Values: "mono" / "stereo"
- bit_rate	integer	Returns audio bitrate (unit in kbps) Value Range: 32 ~ 256
- quality	integer	Returns signal quality level Value Range: 0 - 100
- tune_aid	integer	Returns signal strength level Value Range: 0 - 100
- off_air	boolean	Returns Off Air status
- dab_plus	boolean	Returns DAB+ status
- program_type	string	Returns Program Type
- ch_label	string	Returns CH Label
- service_label	string	Returns Service Label
- dls	string	Returns DLS
- ensemble_label	string	Returns Ensemble Label
- initial_scan_progress	integer	Returns Initial Scan progress status. Available only when "dab_initial_scan" exists in tuner - func_list under /system/getFeatures Value Range: 0 - 100
- total_station_num	integer	Returns station numbers detected by Initial Scan. Available only when "dab_initial_scan" exists in tuner - func_list under /system/getFeatures 0 if Initial Scan hasn't executed or nothing found. Value Range: 0 - 255

Example Response	<pre>{ "response_code":0, "band":"fm", "auto_scan":false, "am":{ "preset":8, "freq":531, "tuned":true }, "fm":{ "preset":0,</pre>
------------------	---

	<pre> "freq":87500, "tuned":true }, "rds":{ "program_type":"Affairs", "program_service":"RDS PS", "radio_text_a":"RDS Radio Text A", "radio_text_b":"RDS Radio Text B" }, "dab":{ "preset":10, "id":10000, "status":"ready", "freq":22564, "category":"primary", "audio_mode":"stereo", "bit_rate":128, "quality":90, "tune_aid":80, "off_air":false, "dab_plus":false, "program_type":"Pop Music", "ch_label":"12D", "service_label":"BBC Radio 1", "dls":"The Chart Game: www.bbc.co.uk/radio2", "ensemble_label":"BBC Network DAB" } } </pre>
--	--

6.3. setBand

For setting Tuner Band.

Request Parameters

name	Type	required	description
band	String	yes	Specifies Band Values: "am" / "fm" / "dab"

URI	<BaseURL>/v1/tuner/setBand?band=<band>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/setBand?band=fm

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre> { "response_code":0 } </pre>
------------------	--------------------------------------

	}
--	---

6.4. setFreq

For setting Tuner frequency.

Request Parameters

name	Type	required	description
band	String	yes	Specifies Band Values : "am" / "fm"
tuning	String	yes	Specifies a tuning method. Use "tp_up" and "tp_down" only when Band is RDS Values: "up" / "down" / "cancel" / "auto_up" / "auto_down" / "tp_up" / "tp_down" / "direct"
num	Integer	no	Specifies frequency (unit in kHz). Valid only when tuning is "direct"

URI	<BaseURL>/v1/tuner/setFreq?band=<band>&tuning=<tuning>&num=<num>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/setFreq?band=fm&tuning=direct&num=87500

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

6.5. recallPreset

For recalling a Tuner preset through preset number.

Request Parameters

name	Type	required	description
zone	String	yes	Specifies station recalling zone. This causes input change in specified zone Values: "main" / "zone2" / "zone3" / "zone4"
band	String	yes	Specifies Band type. Depending on Preset Type gotten via /system/getFeatures, specifying value is different. Values: "common" (band common) / "am" / "fm" / "dab" (separate)
num	Integer	yes	Specifies Preset number Value: one in the range gotten via /system/getFeatures

URI	<BaseURL>/v1/tuner/recallPreset?zone=<zone>&band=<band>&num=<num>
Method	GET

Example Request	http://{host}/YamahaExtendedControl/v1/tuner/recallPreset?zone=main&band=fm&num=13
-----------------	--

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

6.6. switchPreset

For selecting Tuner preset.

Call this API after change the target zone's input to Tuner. It is possible to change Band in case of preset type is “common”. In case of preset type is “separate”, need to change the target Band before calling this API.

This API is available on and after API Version 1.17.

Request Parameters

name	Type	required	description
dir	String	yes	Specifies change direction of preset Values: "next" / "previous"

URI	<BaseURL>/v1/tuner/switchPreset?dir=<dir>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/switchPreset?dir=next		

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

6.7. storePreset

For registering current station to a preset.

Request Parameters

name	Type	required	description
num	Integer	yes	Specifying a preset number Value: one in the range gotten via /system/getFeatures

URI	<BaseURL>/v1/tuner/storePreset?num=<num>		
-----	--	--	--

Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/storePreset?num=10

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

6.8. clearPreset

For clearing Tuner preset.

Request Parameters

name	Type	required	description
band	String	yes	Specifies Band. Values depend on Preset Type gotten via /system/getFeatures Values: "common" (common), "am" / "fm" / "dab" (separate)
num	Integer	yes	Specifies Preset number Values: one in the range gotten via /system/getFeatures

URI	<BaseURL>/v1/tuner/clearPreset?band=<band>&num=<num>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/clearPreset?band=fm&num=10

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to “Response Code List” for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

6.9. startAutoPreset

For starting Auto Preset. Available only when "fm_auto_preset" exists in tuner - func_list under /system/getFeatures.

Request Parameters

name	Type	required	description
band	String	yes	Specifies Band. Value : "fm"

URI	<BaseURL>/v1/tuner/startAutoPreset?band=<band>
-----	--

Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/startAutoPreset?band=fm

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

6.10. cancelAutoPreset

For canceling Auto Preset. Available only when "fm_auto_preset" exists in tuner - func_list under /system/getFeatures.

Request Parameters

name	Type	required	description
band	String	yes	Specifies Band. Value : "fm"

URI	<BaseURL>/v1/tuner/cancelAutoPreset?band=<band>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/cancelAutoPreset?band=fm		

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

6.11. movePreset

For moving preset.

For example, if excute movePreset?from=4&to=2 for list {[A], [B], [C], [D], [E] ...}, list is arranged as {[A], [D], [B], [C], [E] ...}.

Request Parameters

name	Type	required	description
band	String	yes	Specifies Band. Values depend on Preset Type gotten via /system/getFeatures. Values: "common" (common), "am" / "fm" / "dab" (separate)
from	Integer	yes	Specifies source preset number Values: one in the range gotten via /system/getFeatures

to	Integer	yes	Specifies destination preset number Values: one in the range gotten via /system/getFeatures
----	---------	-----	--

URI	<BaseURL>/v1/tuner/movePreset?band=<band>&from=<from>&to=<to>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/movePreset?band=am&from=10&to=3		

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

6.12. startDabInitialScan

For starting DAB Initial Scan. Available only when "dab_initial_scan" exists in tuner - func_list under /system/getFeatures.

Request Parameters

URI	<BaseURL>/v1/tuner/startDabInitialScan		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/startDabInitialScan		

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

6.13. cancelDabInitialScan

For canceling DAB Initial Scan. Available only when "dab_initial_scan" exists in tuner - func_list under /system/getFeatures.

Request Parameters

URI	<BaseURL>/v1/tuner/cancelDabInitialScan		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/cancelDabInitialScan		

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error

		Refer to " Response Code List " for details
Example Response	<pre>{ "response_code":0 }</pre>	

6.14. setDabTuneAid

For executing DAB Tune Aid. Available only when "dab_tune_aid" exists in tuner - func_list under /system/getFeatures.

Request Parameters

name	Type	required	description
action	String	yes	Specifies action of Tune Aid. "stop" / "up" / "down" are available only when executing Tuner Aid. Values : "start" / "stop" / "up" / "down"

URI	<BaseURL>/v1/tuner/setDabTuneAid?action=<action>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/setDabTuneAid?action=up		

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre>{ "response_code":0 }</pre>	
------------------	------------------------------------	--

6.15. setDabService

For selecting DAB Service. Available only when DAB is valid to use.

Request Parameters

name	Type	required	description
dir	String	yes	Specifies change direction of services Values: "next" / "previous"

URI	<BaseURL>/v1/tuner/setDabService?dir=<dir>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/tuner/setDabService?dir=next		

Response Parameters

name	Type	description
response_code	Integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

7. Network/USB

APIs in regard to Network/USB related setting and getting information.

Target Inputs: USB / Network related ones (Server / Net Radio / Pandora / Spotify / AirPlay etc.)

7.1. getPresetInfo

For retrieving preset information. Presets are common use among Net/USB related input sources.

Request Parameters

URI	<BaseURL>/v1/netusb/getPresetInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getPresetInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
preset_info	array	Returns preset info. Element number of an array can be gotten via system/getFeatures
- input	string	Returns Input ID. Returns "unknown" if no presets available Refer to " ALL ID List " for details
- text	string	Returns text info. Returns "" (empty text) if no presets available
func_list	string array	Returns a list of valid functions for Preset. (Recall/Store functions are always valid without specifically listed here) Values: "clear" / "move"

Example Response	<pre>{ "response_code":0, "preset_info":[{ "input":"net_radio", "text":"CBC Radio Jazz" }, { "input":"pandora", "text":"Queen Radio" }, ... { "input":"unknown", "text":"" }], "func_list":["clear",</pre>
------------------	---

	<pre> "move"] } </pre>
--	-------------------------

7.2. getPlayInfo

For retrieving playback information.

Request Parameters

URI	<BaseURL>/v1/netusb/getPlayInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getPlayInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details
input	string	Returns current Net/USB related Input ID Refer to "ALL ID List" for details
playback	string	Returns playback status Values: "play" / "stop" / "pause" / "fast_reverse" / "fast_forward"
repeat	string	Returns repeat setting status Value: "off" / "one" / "all"
shuffle	string	Returns shuffle setting status Values: "off" / "on" / "songs" / "albums"
repeat_available	string array	Returns current settable repeat setting (Available on after API version 1.19) Values: "off" / "one" / "all"
shuffle_available	string array	Returns current settable shuffle setting (Available on after API version 1.19) Values: "off" / "on" / "songs" / "albums"
play_time	integer	Returns current playback time (unit in second). Returns -60000 if playback time is invalid Value Range: -60000 (invalid) / -59999 ~ 59999 (valid)
total_time	integer	Returns total playback time (unit in second). Returns 0 if total time is not available or invalid Value Range: 0 ~ 59999
artist	string	Returns artist name. Returns station name if the input is Net Radio / Pandora / radiko. Returns station name/artist name if the input is Napster (Radio). If Net Radio is airable.radio, "(location / language)" will be appended to the station name. Returns ad name if Pandora playbacks ad contents. If input is MC Link, returns sender(master)'s internal content info or Room Name if the sender(master) input is one of external sources Text information may be left in artist / album / track while playback is stopped. At this time, you can expect playback to start by sending a Play request, but another song different from the displayed song such

		as when playing the station may be played. Please also note that there is no guarantee that playback will resume certainly depending on the situation.
album	string	Returns album name. Returns channel name if the input is SiriusXM. Returns subtitle name if the input is radiko. Returns company name if Pandora playbacks an ad. If input is MC Link, returns sender(master)'s internal content info or Input Name if the sender(master) input is one of external sources
track	string	Returns track name. Returns song name if the input is Napster / SiriusXM / Pandora. Returns title name if the input is radiko. If input is MC Link, returns sender(master)'s internal content info or empty text if the sender(master) input is one of external sources
albumart_url	string	Returns a URL to retrieve album art data. Data is in jpg/png/bmp/ymf format. The path is given as relative address. If "xxx/yyy/zzz.jpg" is returned, the absolute path is expressed as http://{host}/xxx/yyy/zzz.jpg Note: ymf is original format encrypted by Yamaha AV encryption method.
albumart_id	integer	Returns ID to identify album art. If ID got changed, retry to get album art data via albumart_url Value Range: 0 ~ 9999
usb_devicetype	string	Returns USB device type. Returns "unknown" if no USB is connected Values: "msc" / "ipod" / "unknown"
auto_stopped	boolean	Returns whether or not auto top has initiated. If it is true, display appropriate messages to the external application user interface depending on which input current one is. This flag is cleared (set back to false) with these conditions as follows: - Playback is initiated properly - /netusb/setPlayback is executed - type = play found in /netusb/setListControl is executed Target Input : Pandora / SiriusXM A MusicCast Device that detects non-operation time (by key operation on the Device or by remote control) will always return false flag in this data
attribute	integer	Returns playback attribute info. Attributes are expressed as OR of bit field as shown below: b[0] Playable b[1] Capable of Stop b[2] Capable of Pause b[3] Capable of Prev Skip b[4] Capable of Next Skip b[5] Capable of Fast Reverse b[6] Capable of Fast Forward b[7] Capable of Repeat b[8] Capable of Shuffle

		b[9] Feedback Available (Pandora) b[10] Thumbs-Up (Pandora) b[11] Thumbs-Down (Pandora) b[12] Video (USB) b[13] Capable of Bookmark/Favorite (Net Radio / TIDAL / Deezer) b[14] DMR Playback (Server) b[15] Station Playback (Napster) b[16] AD Playback (Pandora) b[17] Shared Station (Pandora) b[18] Capable of Add Track (Napster/Pandora/JUKE/Qobuz) b[19] Capable of Add Album (Napster / JUKE) b[20] Shuffle Station (Pandora) b[21] Capable of Add Channel (Pandora) b[22] Sample Playback (JUKE) b[23] MusicPlay Playback (Server) b[24] Capable of Link Distribution b[25] Capable of Add Playlist (Qobuz) b[26] Capable of add MusicCast Playlist b[27] Capable of Add to Playlist (TIDAL / Deezer) b[28] Seek Backward Available(Spotify) b[29] Seek Forward Available (Spotify) b[30] Seek Available (QQ Music) With Pandora, b[9] = 1 validates "thumbs_up" / "thumbs_down" / "mark_tired" of managePlay and "why_this_song" of getPlayDescription. b[21] = 1 validates "add_channel_track" / "add_channel_artist".
seek_backward_step	integer	Returns the return time (in seconds) for Seek Backward operations. Parameter exists if the attribute Seek Backward Available is set.
seek_forward_step	integer	Returns the return time (in seconds) for Seek Forward operations. Parameter exists if the attribute Seek Forward Available is set.

Example Response	<pre>{ "response_code":0, "input":"usb", "playback":"stop", "repeat":"off", "shuffle":"on", "repeat_available ":["off", "one"], "shuffle_available":["off", "on"], "play_time":200, "total_time":314, "artist":"Bruno Mars",</pre>
------------------	--

	<pre> "album":"24K Magic", "track":"That's What I Like", "albumart_url":"/YamahaRemoteControl/AlbumART/AlbumART.jpg", "albumart_id":18, "usb_devicetype":"msc", "attribute":511 } </pre>
--	---

7.3. setPlayback

For controlling playback status.

Request Parameters

name	type	required	description
playback	string	yes	Specifies playback status Values: "play" / "stop" / "pause" / "play_pause" / "previous" / "next" / "fast_reverse_start" / "fast_reverse_end" / "fast_forward_start" / "fast_forward_end" / "seek_backward" / "seek_forward" / "resume"

URI	<BaseURL>/v1/netusb/setPlayback?playback=<playback>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/setPlayback?playback=stop

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre> { "response_code":0 } </pre>
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7.4. setPlayPosition

For setting track play position. This API is available only when input is Server.

Request Parameters

name	type	required	description
position	integer	yes	Specifies play position (sec) Value : no fewer than total_time gotten via getPlayInfo, nor more than 0

URI	<BaseURL>/v1/netusb/setPlayPosition?position=<position>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/setPlayPosition?position=123

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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7.5. setRepeat

For setting repeat. Available on after API version 1.19.

Request Parameters

name	Type	required	description
mode	string	yes	Specifies the repeat setting Value : "off" / "one" / "all"

URI	<BaseURL>/v1/netusb/setRepeat?mode=<mode>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/setRepeat?mode=off

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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7.6. setShuffle

For setting shuffle. Available on after API version 1.19.

Request Parameters

name	type	required	description
mode	string	yes	Specifies the shuffle setting Value : "off" / "on" / "songs" / "albums"

URI	<BaseURL>/v1/netusb/setShuffle?mode=<mode>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/setShuffle?mode=off

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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7.7. toggleRepeat

For toggling repeat setting. No direct / discrete setting commands available.

Request Parameters

URI	<BaseURL>/v1/netusb/toggleRepeat
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/toggleRepeat

Response Parameters

name	type	Description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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7.8. toggleShuffle

For toggling shuffle setting. No direct / discrete setting commands available.

Request Parameters

URI	<BaseURL>/v1/netusb/toggleShuffle
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/toggleShuffle

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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7.9. getListInfo

For retrieving list information. Basically this info is available to all relevant inputs, not limited to or independent from current input.

Request Parameters

name	type	required	description
list_id	string	no	Specifies list ID. If nothing specified, "main" is chosen implicitly

			Values: "main" (common for all Net/USB sources) "auto_complete" (Pandora) "search_artist" (Pandora) "search_track" (Pandora)
input	string	yes	Specifies target Input ID. Controls for setListControl are to work with the input specified here Values: Input IDs for Net/USB related sources
index	integer	no	Specifies the reference index (offset from the beginning of the list). Note that this index must be in multiple of 8. If nothing was specified, the reference index previously specified would be used Values: 0, 8, 16, 24, ..., 64984, 64992
size	integer	yes	Specifies max list size retrieved at a time Value Range: 1 - 8
lang	string	no	Specifies list language. But menu names or text info are not always necessarily pulled in a language specified here. If nothing specified, English ("en") is used implicitly Values: "en" (English)/ "ja" (Japanese)/ "fr" (French)/ "de" (German)/ "es" (Spanish)/ "ru" (Russian)/ "it" (Italy)/ "zh" (Chinese)

URI	<BaseURL>/v1/netusb/getListInfo?input=<input>&index=<index>&size=<size>&lang=<lang>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getListInfo?input=usb&index=32&size=8&lang=en

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details If codes shown below is returned, display appropriate messages to the external application user interface depending on which input current one is. List info is initialized once an error happens. 100 : Access Error (common for all Net/USB sources) 112 : Access Denied (Server)
input	string	Specifies target Input ID. (Available on and after API Version 1.17)
menu_layer	integer	Returns menu layer Value Range: 0 - 15
max_line	integer	Returns max row number of the list Value Range: 0 - 65000
index	integer	Returns the reference index of the list
playing_index	integer	Returns the index number in a list that is currently in playback. If no list element has playback status, or playback status wasn't found, returns -1
menu_name	string	Returns menu name
list_info	array	Returns list info with the row number size specified by Request Parameter. Returns empty data if index or size wasn't specified. If the max row number is reached in the middle of this list info, the list ends at the max row.

- text	string	Returns text info to display
- subtexts	string array	Returns sub text info. Returns in order of priority to display.
- thumbnail	string	Return a URL to retrieve thumbnail data. Returns "" (empty text) if no sum nail is available
- logo	string	Returns logotype information. If logo is not in need, the value will be "" (Empty string). Values : "hd" (Amazon Music) "ultrahd" (Amazon Music) "hires" (Qobuz)
- attribute	integer	Returns attribute info of list elements. Attributes are expressed as OR of bit field as shown below: b[0] Name exceeds max byte limit (common for all Net/USB sources) b[1] Capable of Select (common for all Net/USB sources) b[2] Capable of Play (common for all Net/USB sources) b[3] Capable of Search (Napster / JUKE) b[4] Album Art available (common for all Net/USB sources) b[5] Now Playing (Pandora) b[6] Capable of Add Bookmark/Favorite (Net Radio / TIDAL / Deezer) b[7] Capable of Add Track (Napster / JUKE / Qobuz) b[8] Capable of Add Album (Napster / JUKE / Qobuz) b[9] Capable of Add Channel (Napster / Pandora) b[10] Capable of Remove Bookmark/Favorite (Net Radio / TIDAL / Deezer) b[11] Capable of Remove Track (Napster / JUKE / Qobuz) b[12] Capable of Remove Album (Napster / JUKE / Qobuz) b[13] Capable of Remove Channel (Napster / Pandora) b[14] Capable of Remove Playlist (Qobuz / TIDAL / Deezer) b[15] Playlist (JUKE / Qobuz) b[16] Radio (JUKE) b[17] Shuffle (Pandora) b[18] Shared Station (Pandora) b[19] Premium Item (radiko) b[20] Capable of Add Artist (Qobuz) b[21] Capable of Remove Artist (Qobuz) b[22] Capable of Add Playlist (Qobuz) b[23] Capable of Play Now b[24] Capable of Play Next b[25] Capable of Add Play Queue b[26] Capable of Add MusicCast Playlist b[27] Capable of Add to Playlist (TIDAL / Deezer) b[28] Capable of Remove from Playlist (TIDAL / Deezer) b[29] Description information available (Qobuz) b[30] Upsell information available (AmazonMusic) Regarding an element with its attributes of b[1] and b[2] both set as 1, it is valid to do both layer movement/shift and start playback by setListControl. If an element got neither b[1] = 0 nor b[2] = 0, it is unavailable content so apply appropriate UI reaction like making it

		gray-out etc.
- appearance	string array	Returns information related to appearance of list element. It is possible for a list element to be set multiple values. Value: "section" (TIDAL / Deezer) "show_more" (TIDAL / Deezer)
upsell	object	Returns information about Upsell Banner. It exists when the bit with Upsell information is set in attribute.
- title	string	Return title text.
- description	string	Returns text of Upsell Banner.

Example Response	<pre>{ "response_code":0, "input":"usb", "menu_layer":3, "max_line":40, "index":32, "playing_index":-1, "menu_name":"The Beatles", "list_info":[{ "text":"Please Please Me", "subtexts":["The Beatles"], "attribute":2, "appearance":[] }, ... { "text":"Let It Be", "subtexts":["The Beatles"], "attribute":20, "appearance":[] }] }</pre>
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7.10. setListControl

For control a list. Controllable list info is not limited to or independent from current input.

Request Parameters

name	type	required	description
list_id	string	no	Specifies list ID. If nothing specified, "main" is chosen implicitly

			Values: "main" (common for all Net/USB sources) "auto_complete" (Pandora) "search_artist" (Pandora) "search_track" (Pandora)
type	string	yes	Specifies list transition type. "select" is to enter and get into one deeper layer than the current layer where the element specified by the index belongs to. "play" is to start playback current index element, "return" is to go back one upper layer than current. "select" and "play" needs to specify an index at the same time. In case to "select" an element with its attribute being "Capable of Search", specify search text using setSearchString in advance. (Or it is possible to specify search text and move layers at the same time by specifying an index in setSearchString) Values: "select" / "play" / "return"
index	integer	no	Specifies an element position in the list being selected (offset from the beginning of the list). This is mandatory to specify if the type is "select" or "play" Value Range: 0 - 64999
zone	string	no	Specifies target zone to playback. In the specified zone, input change occurs at the same time of playback. This parameter is valid only when type "play" is specified. If nothing is specified, "main" is chosen implicitly 値 : "main" / "zone2" / "zone3" / "zone4"

URI	<BaseURL>/v1/netusb/setListControl?list_id=<list_id>&type=<type>&index=<index>&zone=<zone>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/setListControl?list_id=main&type=play&index=50&zone=main

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details

Example Response	{ "response_code":0 }
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7.11.setSearchString

For setting search text. Specifies string executing this API before select an element with its attribute being "Capable of Search" or retrieve info about searching list(Pandora).

Request Parameters

name	type	required	description
list_id	string	no	Specifies list ID. If nothing specified, "main" is chosen implicitly Values : "main" (All) "auto_complete" (Pandora)

			"search_artist" (Pandora) "search_track" (Pandora)
string	string	yes	Setting search text
index	integer	no	Specifies an element position in the list being selected (offset from the beginning of the list). Valid only when the list_id is "main" Specifies index an element with its attribute being "Capable of Search" Controls same as setListControl "select" are to work with the index an element specified. If no index is specified, non-actions of select Values : 0 ~ 64999

URI	<BaseURL>/v1/netusb/setSearchString
Method	POST
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/netusb/setSearchString
Example Request (json)	{ "list_id":"auto_complete", "string":"michael" }

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details

Example Response	{ "response_code":0 }
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7.12. recallPreset

For recalling a content preset.

Request Parameters

name	type	required	description
zone	string	yes	Specifies station recalling zone. This causes input change in specified zone Values: "main" / "zone2" / "zone3" / "zone4"
num	integer	yes	Specifies Preset number Value: one in the range gotten via /system/getFeatures

URI	<BaseURL>/v1/netusb/recallPreset?zone=<zone>&num=<num>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/recallPreset?zone=main&num=13

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error

	Refer to " Response Code List " for details
Example Response	<pre>{ "response_code":0 }</pre>

7.13.storePreset

For registering current content to a preset. Presets are common use among Net/USB related input sources. Confirm the result whether register to a preset or not via netusb - preset_control (11.3 Details of event data).

Request Parameters

name	type	required	description
num	integer	yes	Specifying a preset number Value: one in the range gotten via /system/getFeatures

URI	<BaseURL>/v1/netusb/storePreset?num=<num>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/storePreset?num=10

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre>{ "response_code":0 }</pre>
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7.14.clearPreset

For clearing Preset.

Request Parameters

name	type	required	description
num	integer	yes	Specifies Preset number Values: one in the range gotten via /system/getFeatures

URI	<BaseURL>/v1/netusb/clearPreset?num=<num>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/clearPreset?num=10

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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7.15. movePreset

For moving preset.

For example, if execute movePreset?from=4&to=2 for list {[A], [B], [C], [D], [E] ...}, list is arranged as {[A], [D], [B], [C], [E] ...}.

Request Parameters

name	type	required	description
from	integer	yes	Specifies source preset number Values: one in the range gotten via /system/getFeatures
to	integer	yes	Specifies destination preset number Values: one in the range gotten via /system/getFeatures

URI	<BaseURL>/v1/netusb/movePreset?from=<from>&to=<to>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/movePreset?from=10&to=3

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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7.16. getSettings

For retrieving setup of Net/USB.

Request Parameters

URI	<BaseURL>/v1/netusb/getSettings
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getSettings

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
qobuz	object	Returns information related to Qobuz. Retrievable only when products are equipped with Qobuz functions.
- quality	object	Returns information related to Streaming Quality
- value	string	Returns current Streaming Quality Values: "hr_192_24" (HiRes - 24bits / up to 192kHz)

		"hr_96_24" (HiRes - 24bits / up to 96kHz) "cd_44_16" (CD - 16bits / 44.1kHz) "mp3_320" (MP3 - 320kbps)
- value_list	array	Returns a list of Streaming Quality available
- value	string	Returns value Values: "hr_192_24" / "hr_96_24" / "cd_44_16" / "mp3_320"
- attribute	integer	Returns attribute info. Attributes are expressed as OR of bit field as shown below; b[0] Selectable

Example Response	<pre>{ "response_code":0, "qobuz":{ "quality":{ "value":"hr_192_24", "value_list":[{ "value":"hr_192_24", "attribute":0 }, ... { "value":"mp3_320", "attribute":1 },] } } }</pre>
------------------	---

7.17. setQuality

For setting the reproduction quality of streaming. Refer to available Input/setting value via /system/getSetting.

Request Parameters

name	type	required	description
input	string	yes	Specifies target Input ID Values: "qobuz"
value	string	yes	Specifies reproduction quality. Refer to available value in each input via /netusb/getSettings Values: "hr_192_24" / "hr_96_24" / "cd_44_16" / "mp3_320"

URI	<BaseURL>/v1/netusb/setQuality?input=<input>&value=<value>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/setQuality?input=qobuz&value=hr_192_24

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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7.18. getRecentInfo

For retrieving playback history. History is shared among all Net/USB Input sources.

Request Parameters

URI	<BaseURL>/v1/netusb/getRecentInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getRecentInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
recent_info	array	Returns playback history. Element number of an array can be gotten via system/getFeatures
- input	string	Returns Input ID
- text	string	Returns text information
- albumart_url	string	Returns a URL to retrieve album art data. Returns "" (empty text) if no URL is available.
- play_count	integer	Returns number of playback count. Playback count is cleared when it's cleared in playback history.

Example Response	{ "response_code":0, "recent_info":[{ "input":"net_radio", "text":"CBC Radio Jazz", "albumart_url":"http://www.yamaha.com/001.jpg", "play_count":1 }, { "input":"pandora", "text":"Queen Radio", "albumart_url":"http://www.yamaha.com/002.jpg", "play_count":3 }, ...] "input":"unknown", }
------------------	--

	<pre> "text": "", "albumart_url": "", "play_count": 1 }] } </pre>
--	--

7.19. recallRecentItem

For recalling a content via playback history.

Request Parameters

name	type	required	description
zone	string	yes	Specifies recalling zone. This causes input change in specified zone Values: "main" / "zone2" / "zone3" / "zone4"
num	integer	yes	Specifies number of memorable playback history. Values: one in the range gotten via /system/getFeatures

URI	<BaseURL>/v1/netusb/recallRecentItem?zone=<zone>&num=<num>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/recallRecentItem?zone=main&num=13

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre> { "response_code": 0 } </pre>
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7.20. clearRecentInfo

For clearing all recent history information.

Request Parameters

URI	<BaseURL>/v1/netusb/clearRecentInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/clearRecentInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	<pre> { "response_code": 0 } </pre>
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7.21. managePlay

For special processing for track.

Request Parameters

name	type	required	description
type	string	yes	Specifies type of special processing Vale: "add_bookmark" (Net Radio / TIDAL / Deezer) "add_track" (Napster / Pandora / JUKE / Qobuz) "add_album" (Napster / JUKE) "add_channel_track" (Pandora) "add_channel_artist" (Pandora) "add_playlist" (Qobuz) "add_to_playlist" (TIDAL / Deezer) "thumbs_up" (Pandora) "thumbs_down" (Pandora) "mark_tired" (Pandora)
timeout	integer	yes	Specifies timeout duration(ms) for this API process. If specifies 0, treat as maximum vale Value: 0 ~ 60000

URI	<BaseURL>/v1/netusb/managePlay?type=<type>&timeout=<timeout>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/managePlay?type=add_bookmark&timeout=5000

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details If codes shown below is returned, display appropriate messages to the External application user interface. 100 : Access error (All) 110 : Read Only Mode (Pandora) 111 : Max Stations (Pandora)

Example Response	{ "response_code":0 }
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7.22. manageList

For special processing for list.

Request Parameters

name	type	required	description
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list_id	string	no	Specifies list ID. If nothing specified, "main" is chosen implicitly. Values : "main" (All) "auto_complete" (Pandora) "search_artist" (Pandora) "search_track" (Pandora)
type	string	yes	Specifies type of special processing Value: "add_bookmark" (Net Radio / TIDAL / Deezer) "add_track" (Napster / JUKE / Qobuz) "add_album" (Napster / JUKE / Qobuz) "add_artist" (Qobuz) "add_channel" (Napster / Pandora) "add_playlist" (Qobuz) "remove_bookmark" (Net Radio / TIDAL / Deezer) "remove_track" (Napster / JUKE / Qobuz) "remove_album" (Napster / JUKE / Qobuz) "remove_artist" (Qobuz) "remove_channel" (Napster / Pandora) "remove_playlist" (Napster / Qobuz) "remove_from_playlist" (TIDAL / Deezer) "end_auto_complete" (Pandora) "end_auto_complete" must be excuted when make a Auto Complete window (Pandora) disappear. (certification requirement)
index	integer	no	Specifies the reference index (offset from the beginning of the list). No parameter set if type is "end_auto_complete" Vaule: 0 ~ 64999
zone	string	no	Specifies recalling zone. This causes input change in the specified zone. This parameter is valid only on cases where the list_id is either "auto_complete" / "search_artist" / "search_track", and the type is "add_channel". If no parameter is specified, "main" is chosen implicitly. Values: "main" / "zone2" / "zone3" / "zone4"
timeout	integer	yes	Specifies timeout duration(ms) for this API process. If specifies 0, treat as maximum vale Value: 0 ~ 60000

URI	<BaseURL>/v1/netusb/manageList?list_id=<list_id>&type=<type>&index=<index>&zone=<zone>&bank=<bank>&timeout=<timeout>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/manageList?list_id=main&type=add_bookmark&index=50&timeout=5000

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

		If codes shown below is returned, display appropriate messages to the External application user interface. 100 : Access error (All) 110 : Read Only Mode (Pandora) 111 : Max Stations (Pandora)
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Example Response	<pre>{ "response_code":0 }</pre>
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7.23. getPlayDescription

For retrieving track's detail information.

Request Parameters

name	type	required	description
type	string	yes	Specifies type of special processing Value : "why_this_song" (Pandora)
timeout	integer	yes	Specifies timeout duration(ms) for this API process. If specifies 0, treat as maximum value Value: 0 ~ 60000

URI	<BaseURL>/v1/netusb/getPlayDescription?type=<type>&timeout=<timeout>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getPlayDescription?type=why_this_song&timeout=5000

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
description	string	Returns detail information

Example Response	<pre>{ "response_code":0, "description":"Based on what you've told us so far, we're playing Sanctuary (Closing) by Utad because it features mellow rock instrumentation, electronica influences, interweaving vocal harmony, extensive vamping and acoustic rhythm piano." }</pre>
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7.24. getListDescription

For retrieving list description's detail information.

Request Parameters

name	type	required	description
list_id	string	no	Specifies the list id. If not specified, "main" is treated as specified.

			Vaule: "main" (All)
index	integer	yes	Specifies the reference index (offset from the beginning of the list). Vaule: 0 - 64999
timeout	integer	yes	Specifies timeout duration(ms) for this API process. If specifies 0, treat as maximum value. Vaule: 0 - 60000

URI	<BaseURL>/v1/netusb/getListDescription?list_id=<list_id>&index=<index>&timeout=<timeout>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getListDescription?list_id=main&index=50&timeout=5000		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
Description	string	Specifies detail information
Metadata	object	Specifies metadata that could be acquired
- label	string	Specifies label name
- duration	integer	Specifies song length [sec]
- release	integer	Specifies release date [unix time = time elapsed since January 1, 1970]

Example Response	<pre>{ "response_code":0, "description":"Sample description.", "metadata":{ "label":"Sample Label", "duration":2096, "release":1532037600 } }</pre>
------------------	---

7.25. setListSortOption

For setting List sorting method. Retrieve List information via /netusb/getListInfo after setting.

Request Parameters

name	type	required	description
input	string	yes	Specifies target Input ID. Controls for setListControl are to work with the input specified here Value : "pandora"
type	string	yes	Specifies sorting method Value : values gotten via /system/getFeatures "sort_option_list" If Input is Pandora and there's no "sort_option_list", only "date" / "alphabet" is selectable.

URI	<BaseURL>/v1/netusb/setListSortOption?input=<input>&type=<type>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/setListSortOption?input=pandora&type=date

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
------------------	-----------------------------

7.26. getAccountStatus

For retrieving account information registered on Device.

Request Parameters

URI	<BaseURL>/v1/netusb/getAccountStatus
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getAccountStatus

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
service_list	array	Returns a list of Streaming Service
- id	string	Returns current Input ID Refer to " ALL ID List " for details
- registered	boolean	Returns whether account registered on device correctly. If input is Pandora, returns "false" if no registered account or not select to use registered account.
- login_status	string	Returns login status Value : "logged_in" (All) "logged_out" (All) "access_error" (All) "password_changed" (Napster / SiriusXM) "account_expired" (Napster / JUKE) "invalid_account" (Napster / Spotify) "suspended_account" (Pandora / Spotify)
- type	string	Returns account type Value: "formal" (official account) / "trial" (trial account) / "unpaid" (unpaid account) / "expired" (expired account)
- trial_time_left	integer	Returns remaining trial days only account type is "trial". If this value is 0, trial account is set to expire within 24 hours. Return -1 if trial account out of date. Return -2 if not retrieving information from server.

Example Response	<pre> { "response_code":0, "service_list":[{ "id":"napster", "registered":true, "login_status":"logged_in", "username":"", "type":"formal", "trial_time_left":0 }, { "id":"siriusxm", "registered":true, "login_status":"logged_in", "username":"", "type":"trial", "trial_time_left":7 }, ... { "id":"qobuz", "registered":false, "login_status":"logged_out", "username":"", "type":"formal", "trial_time_left":0 }] }</pre>
------------------	--

7.27. getServiceInfo

For retrieving information of various Streaming Service. The combination of Input/Type is available as follows;

Input	Type
Pandora	Account List (account_list) : retrieving list of account registred on Device Licensing (licensing) : checking license Activation Code (activation_code) : retrieving Activation Code ※Disable to check Napster license by refering the value of this APIs response_code. Device issues events of netusb - account_updated by condition, retrieve the info excute /netusb/getAccountStatus. (Sometimes Device not issue events) ※Before retrieve Activation Code, retrieve Account List and check not to reach Max about the num of registration.

Request Parameters

name	type	required	description
input	string	yes	Specifies target Input ID. Value: "pandora" / "napster"
type	string	yes	Specifies type of retrieving info Value: "account_list" (Pandora) "licensing" (Napster / Pandora) "activation_code" (Pandora)
timeout	integer	yes	Specifies timeout duration(ms) for this API process. If specifies 0, treat as maximum value. Value: 0 ~ 60000

URI	<BaseURL>/v1/netusb/getServiceInfo?input=<input>&type=<type>&timeout=<timeout>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getServiceInfo?input=pandora&type=account_list&timeout=10000

Response Parameters (Pandora – Account List)

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
account_list	array	Returns list of account
- index	integer	Returns index to discem account Value: 0 ~ 7
- selected	boolean	Returns whether this account is selected

Example Response	<pre>{ "response_code":0, "account_list":[{ "index":0, "username": "", "selected":true }, ... { "index":7, "username": "", "selected":false }] }</pre>
------------------	--

Response Parameters (Pandora – Licensing)

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

		If codes shown below is returned, display appropriate messages to the external application user interface. 100 : Access error 109 : License error
--	--	--

Example Response	<pre>{ "response_code":0 }</pre>
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Response Parameters (Napster – Licensing)

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details Note that disable to check license by referring the value of response_code.

Example Response	<pre>{ "response_code":0 }</pre>
------------------	------------------------------------

Response Parameters (Pandora – Activation Code)

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to “Response Code List” for details If codes shown below is returned, display appropriate messages to the external application user interface. 100 : Access error 106 : reach account num ceiling 110 : Read Only Mode
url	string	Returns Activation URL
code	string	Returns Activation Code
index	integer	Returns index to discem account. Value: 0 ~ 7

Example Response	<pre>{ "response_code":0, "url":"http://www.pandora.com/yamaha", "code":"00000000", "index":0 }</pre>
------------------	---

7.28. getPlayErrorInfo

Acquires information about playback errors.

It can be acquired when the following playback error is notified in the event.

- 12 : Device Conflicted

Request Parameters

URI	<BaseURL>/v1/netusb/getPlayErrorInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/netusb/getPlayErrorInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details
error	array	Returns information about playback errors.
- type	string	Returns the error type. Value: "device_conflicted"
- title	string	Returns title of the error.
- description	string	Returns a detailed description of the error.

Example Response	<pre>{ "response_code":0, "error":[{ "type":"device_conflicted", "title":"device conflicted", "description":"Do you want to play on this device?" }] }</pre>
------------------	--

8. CD

APIs in regard to CD setting and getting information.

8.1. getPlayInfo

For retrieving playback information of CD.

Request Parameters

URI	<BaseURL>/v1/cd/getPlayInfo
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/cd/getPlayInfo

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details
device_status	string	Returns current CD device status Values: "open" / "close" / "ready" / "not_ready"
playback	string	Returns playback status Values: "play" / "stop" / "pause" / "fast_reverse" / "fast_forward"
repeat	string	Returns repeat setting status Value: "off" / "one" / "all" / "folder" / "a-b"
shuffle	string	Returns shuffle setting status Values: "off" / "on" / "folder" / "program"
repeat_available	string array	Returns current settable repeat setting (Available on after API version 1.19) Values: "off" / "one" / "all" / "folder"
shuffle_available	string array	Returns current settable shuffle setting (Available on after API version 1.19) Values: "off" / "on" / "folder"
play_time	integer	Returns current playback time (unit in second). Value Range: -59999 ~ 59999
total_time	integer	Returns current track total playback time (unit in second). Value Range: 0 ~ 59999
disc_time	integer	Returns CD total playback time (unit in second). Value Range: 0 ~ 59999
track_number	integer	Returns track number that is currently in playback. If no track, or playback status is complete stop, returns -1. Values: 0 ~ 512
total_tracks	integer	Returns total track number Values: 0 ~ 512
artist	string	Returns artist name.
album	string	Returns album name.
track	string	Returns track name.

Example Response	<pre>{ "response_code":0, "device_status":"ready", "playback":"play", "repeat":"all", "shuffle":"on", "repeat_available":["off", "one", "all", "folder"], "shuffle_available":["off", "on", "folder"], "play_time":100, "total_time":300, "track_number":5, "total_tracks":13, "artist":"Mr.Children", "album":"SUPERMARKET FANTASY", "track":"GIFT" }</pre>
------------------	--

8.2. setPlayback

For controlling playback status.

Request Parameters

name	type	required	description
playback	string	yes	Specifies playback status Values: "play" / "stop" / "pause" / "previous" / "next" / "fast_reverse_start" / "fast_reverse_end" / "fast_forward_start" / "fast_forward_end" / "track_select"
num	integer	no	Specifies target track number to playback. This parameter is valid only when playback "track_select" is specified. Values: 1 ~ 512

URI	<BaseURL>/v1/cd/setPlayback?playback=<playback>&num=<num>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/cd/setPlayback?playback=stop

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details

Example Response	{ "response_code":0 }
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8.3. toggleTray

For toggling CD tray Open/Close setting.

Request Parameters

URI	<BaseURL>/v1/cd/toggleTray
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/cd/toggleTray

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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8.4. setRepeat

For setting repeat. Available on after API version 1.19.

Request Parameters

name	type	required	description
mode	string	yes	Specifies repeat Values: "off" / "one" / "all" / "folder"

URI	<BaseURL>/v1/cd/setRepeat?mode=<mode>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/cd/setRepeat?mode=off

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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8.5. setShuffle

For setting shuffle. Available on after API version 1.19.

Request Parameters

name	type	required	description
mode	string	yes	Specifies shuffle Values: "off" / "on" / "folder"

URI	<BaseURL>/v1/cd/setShuffle?mode=<mode>		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/cd/setShuffle?mode=off		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }	
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8.6. toggleRepeat

For toggling repeat setting. No direct / discrete setting commands available

Request Parameters

URI	<BaseURL>/v1/cd/toggleRepeat		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/cd/toggleRepeat		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }	
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8.7. toggleShuffle

For toggling shuffle setting. No direct / discrete setting commands available

Request Parameters

URI	<BaseURL>/v1/cd/toggleShuffle		
Method	GET		
Example Request	http://{host}/YamahaExtendedControl/v1/cd/toggleShuffle		

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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9. Clock

APIs in regard to Clock setting and getting information.

9.1. getSettings

For retrieving setting related to Clock function.

Request Parameters

URI	<BaseURL>/v1/clock/getSettings
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/clock/getSettings

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details
auto_sync	boolean	Returns clock time auto sync setting
format	string	Returns format of time display Values : "12h" (12-hour notation) / "24h" (24-hour notation)
alarm	object	Returns information related to alarm function
- alarm_on	boolean	Returns alarm function status on/off
- volume	integer	Returns alarm volume value Vaule Range : calculated by minimum/maximum/step value gotten via /system/getFeatures "alarm_volume"
- fade_interval	integer	Returns alarm fade interval (unit in second) Vaule Range : calculated by minimum/maximum/step value gotten via /system/getFeatures "alarm_fade"
- fade_type	integer	Returns alarm fade type Value : one in the range gotten via /system/getFeatures "alarm_fade_type_num"
- mode	string	Returns alarm mode Value : one gotten via /system/getFeatures "alarm_mode_list"
- repeat	boolean	Returns repeat setting. This parameter is valid only when alarm mode "oneday" is specified
- oneday	object	Returns detail information about alarm mode "oneday". This parameter is valid only when "oneday " exists in alarm mode
- enable	boolean	Returns whether or not "oneday" setup is valid
- time	string	Returns alarm start-up time. Format is "hhmm" Values : hh : 00 ~ 23 (Hour) mm : 00 ~ 59 (Minute)
- beep	boolean	Returns whether or not beep is valid
- playback_type	string	Returns playback type Value : "resume" / "preset"
- resume	object	Returns detail infomarion when resume is specified. This parameter is valid only when playback_type "resume" is specified.
- input	string	Returns target Input ID to playback

		Values : Input IDs gotten via /system/getFeatures "alarm_input_list"
- preset	object	Returns detail information when preset is specified. This parameter is valid only when playback_type "preset" is specified.
- type	string	Returns preset type Values : Type gotten via /system/getFeatures "alarm_preset_list"
- num	integer	Returns preset number. Selectable preset number in each preset type is readable in /system/getFeatures.
- netusb_info	object	Returns preset information related to Net/USB input sources. This parameter is valid only when preset type "netusb" is specified.
- input	string	Returns Input ID. "unknown" when no registrations.
- text	string	Returns text information. "" (empty text) when no registrations.
- tuner_info	object	Returns preset information related to Tuner. This parameter is valid only when preset type "tuner_common" / "am" / "fm" / "dab"
- band	string	Returns Band information. "unknown" when no presets. Value : "am" / "fm" / "dab" / "unknown"
- number	integer	Returns; frequency (unit in kHz) (band = AM or FM) Station ID (band = DAB) 0 when there's no presets
- snooze	boolean	Returns snooze setting. Available only when "snooze" exists in func_list under /system/getFeatures.
- sunday	object	Returns sunday setup information when alarm mode "weekly". This parameter is valid only when " weekly " exists in alarm mode. Data format is same as alarm mode "oneday" above.
- monday	object	Returns monday setup information when alarm mode "weekly". This parameter is valid only when " weekly " exists in alarm mode. Data format is same as alarm mode "oneday" above.
- tuesday	object	Returns tuesday setup information when alarm mode "weekly". This parameter is valid only when " weekly " exists in alarm mode. Data format is same as alarm mode "oneday" above.
- wednesday	object	Returns wednesday setup information when alarm mode "weekly". This parameter is valid only when " weekly " exists in alarm mode. Data format is same as alarm mode "oneday" above.
- thursday	object	Returns thursday setup information when alarm mode "weekly". This parameter is valid only when " weekly " exists in alarm mode. Data format is same as alarm mode "oneday" above.
- friday	object	Returns friday setup information when alarm mode "weekly". This parameter is valid only when " weekly " exists in alarm mode. Data format is same as alarm mode "oneday" above.
- saturday	object	Returns Saturday setup information when alarm mode "weekly". This parameter is valid only when " weekly " exists in alarm mode. Data format is same as alarm mode "oneday" above.

Example Response	{ "response_code":0, "auto_sync":true, "alarm":{ "alarm_on":true,
------------------	---

	<pre> "volume":40, "fade_interval":180, "fade_type":1, "mode":"oneday", "repeat":false, "oneday":{ "enable":true, "time":"0730", "beep":false, "playback_type":"preset", "preset":{ "type":"netusb", "num":8, "netusb_info":{ "input":"pandora", "text":"Queen Radio" } } }, ... "saturday":{ ... } } </pre>
--	--

9.2. setAutoSync

For setting clock time auto sync. Available only when "date_and_time" exists in clock - func_list under /system/getFeatures.

Request Parameters

name	type	required	description
enable	boolean	yes	Specifies whether or not clock auto sync is valid

URI	<BaseURL>/v1/clock/setAutoSync?enable=<enable>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/clock/setAutoSync?enable=true

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details

Example Response	<pre> { "response_code":0 } </pre>
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9.3. setDateAndTime

For setting date and clock time. Available only when "date_and_time" exists in clock - func_list under /system/getFeatures.

Request Parameters

name	type	required	description
date_time	string	yes	Specifies date and time set on device. Format is "YYMMDDhhmmss". Value : YY : 00 ~ 99 (Year / 2000 ~ 2099) MM : 01 ~ 12 (Month) DD : 01 ~ 31 (Day) hh : 00 ~ 23 (Hour) mm : 00 ~ 59 (Minute) ss : 00 ~ 59 (Second)

URI	<BaseURL>/v1/clock/setDateAndTime?date_time=<date_time>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/clock/setDateAndTime?date_time=150610224530

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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9.4. setClockFormat

For setting format of time display. Available only when " clock_format " exists in clock - func_list under /system/getFeatures.

Request Parameters

name	type	required	description
format	string	yes	Returns format of time display Values : "12h" (12-hour notation) / "24h" (24-hour notation)

URI	<BaseURL>/v1/clock/setClockFormat?format=<format>
Method	GET
Example Request	http://{host}/YamahaExtendedControl/v1/clock/setClockFormat?format=24h

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to " Response Code List " for details

Example Response	{ "response_code":0 }
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9.5. setAlarmSettings

For setting alarm function.

Request Parameters

name	type	required	description
alarm_on	boolean	no	Specifies alarm function status on/off
volume	integer	no	Specifies alarm volume value Value Range : calculated by minimum/maximum/step value gotten via /system/getFeatures "alarm_volume"
fade_interval	integer	no	Specifies alarm fade interval (unit in second) Value Range : calculated by minimum/maximum/step value gotten via /system/getFeatures "alarm_fade"
fade_type	integer	no	Specifies alarm fade type Value : 1 ~ fade_type_max (value gotten via /system/getFeatures)
mode	string	no	Specifies alarm mode Value : one gotten via /system/getFeatures "alarm_mode_list"
repeat	boolean	no	Specifies repeat setting. This parameter is valid only when alarm mode "oneday" is specified
detail	object	no	Setting detail information of alarm function
- day	string	no	Specifies target date for alarm setting. This parameter is specified certainly when set detail parameters. Value : "oneday" / "sunday" / "monday" / "tuesday" / "wednesday" / "thursday" / "friday" / "saturday"
- enable	boolean	no	Specifies valid / invalid of the alarm setting of the target date.
- time	string	no	Specifies alarm start-up time. Format is "hhmm" Values : hh : 00 ~ 23 (Hour) mm : 00 ~ 59 (Minute)
- beep	boolean	no	Specifies whether or not beep is valid.
- playback_type	string	no	Specifies playback type Value : "resume" / "preset"
- resume	object	no	Specifies detail information when resume is specified. This parameter is valid only when playback_type "resume" is specified.
- input	string	no	Specifies target Input ID to playback. No playback when "none" is specified. Values : Input IDs gotten via /system/getFeatures "alarm_input_list"
- preset	object	no	Specifies detail information when preset is specified. This parameter is valid only when playback_type "preset" is specified.
- type	string	no	Specifies preset type Values : Type gotten via /system/getFeatures "alarm_preset_list"
- num	integer	no	Specifies preset number. Selectable preset number in each preset

			type is readable in /system/getFeatures.
- snooze	boolean	no	Returns snooze setting. Available only when "snooze" exists in func_list under /system/getFeatures.

URI	<BaseURL>/v1/clock/setAlarmSettings
Method	POST
Example Request (URI)	http://{host}/YamahaExtendedControl/v1/clock/setAlarmSettings
Example Request (json)	<pre>{ "alarm_on":true, "volume":40, "fade_interval":180, "fade_type":1, "mode":"oneday", "repeat":false, "detail":{ "day":"oneday", "enable":true, "time":"0730", "beep":false, "playback_type":"preset", "preset":{ "type":"netusb", "num":8 } } }</pre>

Response Parameters

name	type	description
response_code	integer	0: Successful request / Others: Error Refer to "Response Code List" for details

Example Response	<pre>{ "response_code":0 }</pre>
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10. Response Code List

In a response contains one of Response Codes listed below. No other data is included to a response if its response code was other than '0' (Successful request).

code	description
0	Successful request
1	Initializing
2	Internal Error
3	Invalid Request (A method did not exist, a method wasn't appropriate etc.)
4	Invalid Parameter (Out of range, invalid characters etc.)
5	Guarded (Unable to setup in current status etc.)
6	Time Out
99	Firmware Updating
	(100s are Streaming Service related errors)
100	Access Error
101	Other Errors
102	Wrong User Name
103	Wrong Password
104	Account Expired
105	Account Disconnected/Gone Off/Shut Down
106	Account Number Reached to the Limit
107	Server Maintenance
108	Invalid Account
109	License Error
110	Read Only Mode
111	Max Stations
112	Access Denied
113	There is a need to specify the additional destination Playlist
114	There is a need to create a new Playlist
115	Simultaneous logins has reached the upper limit
116	Streaming service disabled
	(200s are distribution related errors)
200	Linking in progress
201	Unlinking in progress

11. Events

11.1. Outline

Events are to notify Devices' status/setup changes immediately to external applications. Events are spread out as UDP unicast.

11.2. Conditions

When a request header from an external application contains fields shown below, a Device issues events of status changes. If no event is required, do not include the specified fields:

X-AppName:MusicCast/XXX(YYY)

X-AppPort:ZZZ

*Here XXX stands for an application version, YYY shows OS, ZZZ shows event receiving port.

Following is an example of MusicCast CONTROLLER:

X-AppName:MusicCast/1.40(iOS)

X-AppPort:41100

Event notification timeouts in 10 minutes if no further event request is sent from an IP address set as event receiving device. If another request is made within 10 minutes of previous request, the timeout duration is reset and extended.

Event receiving port will be overwritten if a different port number is sent as a request by the registered device using X-AppPort.

11.3. Details of event data

Following data sets only which had status/setting change and which are valid on each Device and each zone are to be notified by events.

Event Parameters

name	type	Description
system	object	Returns System related information
- bluetooth_info_updated	boolean	Returns whether or not Bluetooth info has changed. If so, pull renewed info using /system/getBluetoothInfo
- bluetooth_device_list_updated	boolean	Returns whether or not Bluetooth (sink) device list has changed. If so, pull renewed info using /system/getBluetoothDeviceList
- func_status_updated	boolean	Returns whether or not System overall info has changed. If so, pull renewed info using /system/getFuncStatus
- name_text_updated	boolean	Returns whether or not name text info has changed. If so, pull renewed info using /system/getNameText .
- location_info_updated	boolean	Returns whether or not Location info has changed. If so, pull renewed info using /system/getLocationInfo

main	object	Returns Main Zone related information
- power	string	Returns power status Values: "on" / "standby"
- input	string	Returns current Input ID Values: Input IDs gotten via /system/getFeature
- volume	integer	Returns volume value Values: Value range calculated by minimum/maximum/step values gotten via /system/getFeatures
- mute	boolean	Returns mute status
- status_updated	boolean	Returns whether or not other info has changed than main zone power/input/volume/mute status. If so, pull renewed info using /main/getStatus
- signal_info_updated	boolean	Returns whether or not signal info has changed. If so, pull renewed info using /main/getSignalInfo
zone2	object	Returns information related to Zone2. This data format is same as main zone's one above.
zone3	object	Returns information related to Zone3. This data format is same as main zone's one above.
zone4	object	Returns information related to Zone4. This data format is same as main zone's one above.
tuner	object	Returns Tuner related information
- play_info_updated	boolean	Returns whether or not playback info has changed. If so, pull renewed info using /tuner/getPlayInfo
- preset_info_updated	boolean	Returns whether or not preset info has changed. If so, pull renewed info using /tuner/getPresetInfo
netusb	object	Returns Net/USB related information
- play_error	integer	Returns error codes happened during playback for displaying appropriate messages to the external application user interface. If multiple errors happen at the same time, refer to the value of multiple_play_errors sent together for proper messaging. In the case of Device Conflicted (12), refer to /netusb/getPlayErrorInfo for detailed information and display the message. Values: 0: No Error 1: Access Error (common for all Net/USB sources) 2: Playback Unavailable (common for all Net/USB sources) 3: Skip Limit Reached (Napster / Pandora) 4: Invalid Session (Napster / SiriusXM) 5: High-Resolution File Not Playable at MusicCast Leaf (Server) 6: User Uncredentialed (Qobuz) 7: Track Restricted by Right Holders (Qobuz) 8: Sample Restricted by Right Holders (Qobuz) 9: Genre Restricted by Streaming Credentials (Qobuz) 10: Application Restricted by Streaming Credentials (Qobuz) 11: Intent Restricted by Streaming Credentials (Qobuz) 12 : Device Conflicted (AmazonMusic) 100: Multiple Errors (common for all Net/USB sources)
- multiple_play_errors	integer	Returns bit field flags of multiple playback errors. Flags are expressed as OR of bit field.

		play_error code x is stored as a flag in b[x] shown below. x=0 is reserved for it is for No Error, and x=100 is ignored here b[0] reserved (for it's No Error) b[1] Access Error (common for all Net/USB sources) ... b[11] Intent Restricted by Streaming Credentials (Qobuz)
- play_message	string	Returns playback related message. Max size is 256 bytes
- account_updated	boolean	Returns whether or not account info has changed. If so, pull renewed info using /netusb/getAccountStatus
- play_time	integer	Returns current playback time (unit in second) Value: ranges -59999 - 59999
- preset_info_updated	boolean	Returns whether or not preset info has changed. If so, pull renewed info using netusb/getPresetInfo
- recent_info_updated	boolean	Returns whether or not playback history info has changed. If so, pull renewed info using /netusb/getRecentInfo
- preset_control	object	Returns results of Preset operations
- type	string	Returns a type of Preset operations Values: "store" / "clear" / "recall"
- num	integer	Returns a Preset number being operated Value: one in the range gotten via /system/getFeatures
- result	string	Returns the result of operation Values: "success" (for all types) / "error" (for all types) / "empty" (only for recall) / "not_found" (only for recall)
- trial_status	object	Returns trial status of a Device
- input	string	Returns Input IDs related to Net/USB
- enable	boolean	Returns whether or not trial can be initiated. If false, new trial cannot get started due to a Device in trial status
- trial_time_left	object	Returns remaining time of a trial
- input	string	Returns Net/USB related Input IDs
- time	integer	Returns remaining days of trial. 0 means it expires within 24 hours. -1 means it has expired, -2 means no info is retrieved yet from the server
- play_info_updated	boolean	Returns whether or not playback info has changed. If so, pull renewed info using /netusb/getPlayInfo
- list_info_updated	boolean	Returns whether or not list info has changed. If so, pull renewed info using /netusb/getListInfo
cd	object	Returns CD related information
- device_status	string	Returns CD device status Values: "open" / "close" / "ready" / "not_ready"
- play_time	integer	Returns current playback time (unit in second). Value Range: -59999 ~ 59999
- play_info_updated	boolean	Returns whether or not playback info has changed. If so, pull renewed info using /cd/getPlayInfo
dist	object	Returns Link distribution related information
- dist_info_updated	boolean	Returns whether or not Link distribution Device info has changed. If so, pull renewed info using /dist/getDistributionInfo
- mc_surround	object	Returns MusicCast Surround related information
- control	object	Returns MusicCast Surround operation related information

- type	string	Returns type of operation Values : "build" / "delete"
- result	string	Returns result of operation Values : "success" / "error" / "ch_overlap"
clock	object	Returns Clock related information
- settings_updated	boolean	Returns whether or not clock info has changed. If so, pull renewed info using /clock/getSettings
device_id	string	Return s Device ID.ID is same value using /system/getDeviceInfo Note: Available on and after API Version 1.17, the Device ID is always included in the notification data.

Event Example	<pre> { "system":{ "name_text_updated":true }, "main":{ "power":"on", "input":"siriusxm", "volume":30, "mute":false, "status_updated":true }, "zone2":{ "power":"on", "input":"cd", "volume":50, "mute":false "enhancer":false }, "tuner":{ "play_info_updated":false }, "netusb":{ "play_error ":0, "account_updated":true, "play_time":50, "trial_status":{ "input":"siriusxm", "enable":false }, "trial_time_left":{ "input":"siriusxm", "time":5 }, "play_info_updated":false, "list_info_updated":false }, "cd":{ "tray_status":"ready", </pre>
---------------	--

	<pre>"play_time":100, "play_info_updated":false } }</pre>
--	---

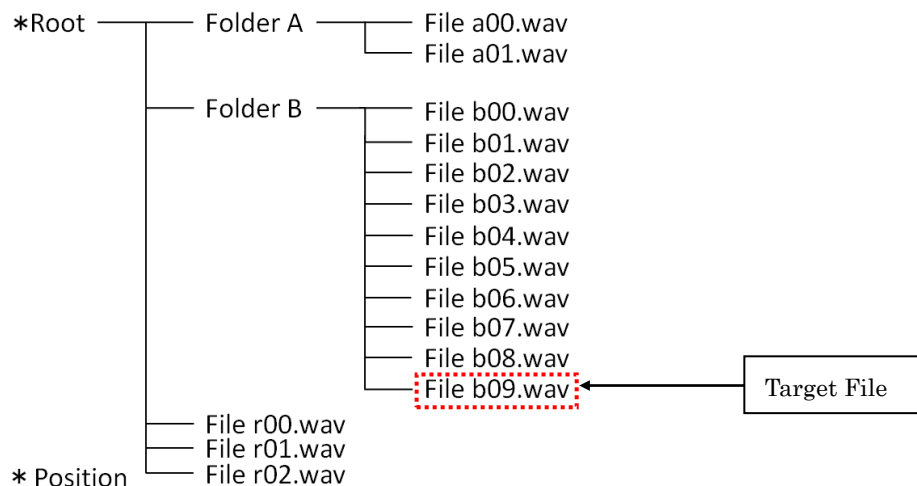
12. All ID List

Type	ID
Zone	main / zone2 / zone3 / zone4 ※Zone A is handles as "main", Zone B is handles as "zone2"
Input	cd / tuner / multi_ch / phono / hdmi1 / hdmi2 / hdmi3 / hdmi4 / hdmi5 / hdmi6 / hdmi7 / hdmi8 / hdmi / av1 / av2 / av3 / av4 / av5 / av6 / av7 / v_aux / aux1 / aux2 / aux / audio1 / audio2 / audio3 / audio4 / audio5 / audio_cd / audio / optical1 / optical2 / optical / coaxial1 / coaxial2 / coaxial / digital1 / digital2 / digital / line1 / line2 / line3 / line_cd / analog / tv / bd_dvd / usb_dac / usb / bluetooth / server / net_radio / napster / pandora / siriusxm / spotify / juke / airplay / radiko / qobuz / tidal / deezer / amazon_music / alexa / qq_music / mc_link / main_sync / none
Sound Program	munich_a / munich_b / munich / frankfurt / stuttgart / vienna / amsterdam / usa_a / usa_b / tokyo / freiburg / royaumont / chamber / concert / village_gate / village_vanguard / warehouse_loft / cellar_club / jazz_club / roxy_theatre / bottom_line / arena / sports / action_game / roleplaying_game / game / music_video / music / recital_opera / pavilion / disco / standard / spectacle / sci-fi / adventure / drama / talk_show / tv_program / mono_movie / movie / enhanced / 2ch_stereo / 5ch_stereo / 7ch_stereo / 9ch_stereo / 11ch_stereo / stereo / all_ch_stereo / surr_decoder / my_surround / target / bass_booster / straight / off

13. Application Notes

13.1. Example of List Control

This shows an example when you change a Device's input to USB, browse a USB memory which has folders/files as shown below and playback the target file "File b09.wav" in the Folder B.



13.1.1. Preparation before start browsing

Send "prepare_input_change" to a Device in order to let it do necessary process before the input change.

Request

`http://{host}/YamahaExtendedControl/v1/main/prepareInputChange?input=usb`

Response

```
{
  "response_code":0
}
```

If input change is expressly necessary, send "setInput" to do so.

Request

`http://{host}/YamahaExtendedControl/v1/main/setInput?input=usb&mode=autoplay_disabled`

Response

```
{
  "response_code":0
}
```

13.1.2. Retrieving list information (In case max line number is less than or equal to 8)

Use "getListInfo" command to get list information.

Here shows how to get it at the root layer.

Request

http://{host}/YamahaExtendedControl/v1/netusb/getListInfo?input=usb&index=0&size=8&lang=en

Response

```
{
  "response_code":0,
  "menu_layer":0,
  "max_line":5,
  "index":0,
  "playing_index":-1,
  "menu_name":"USB",
  "list_info":[
    {
      "text":"FolderA",
      "thumbnail":"",
      "attribute":2
    },
    {
      "text":"FolderB",
      "thumbnail":"",
      "attribute":2
    },
    {
      "text":"File r00",
      "thumbnail":"",
      "attribute":4
    },
    {
      "text":"File r01",
      "thumbnail":"",
      "attribute":4
    },
    {
      "text":"File r02",
      "thumbnail":"",
      "attribute":4
    }
  ]
}
```

← Capability bit[1]: Select bit equal to 1

13.1.3. Layer change

Use “setListControl” to move to Folder B layer. Confirm if the attribute of Response Parameter is selectable as described before.

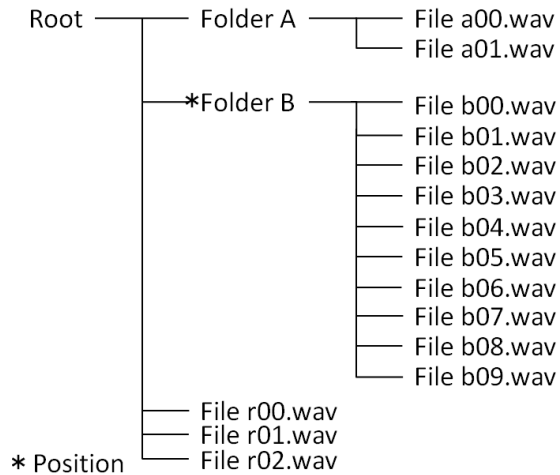
Request

http://{host}/YamahaExtendedControl/v1/netusb/setListControl?list_id=main&type=select&index=1

Response


```
{
  "response_code":0
}
```

Now the current position is moved to Folder B.



13.1.4. Retrieving list information (In case max line number is more than 8)

Use “getListInfo” to get list information.

Here shows how to get it of Folder B.

Request

`http://{host}/YamahaExtendedControl/v1/netusb/getListInfo?input=usb&index=0&size=8&lang=en`

Response

```
{
  "response_code":0,
  "menu_layer":1,
  "max_line": 10 ← Max line number = 10
  "index":0,
  "playing_index":-1,
  "menu_name":"Folder B",
  "list_info":[
    {
      "text":"File b00",
      "thumbnail":"",
      "attribute":4
    },
    {
      "text":"File b01",
      "thumbnail":"",
      "attribute":4
    },
    {
      "text":"File b02",
      "thumbnail":"",

```

```
        "attribute":4
      }
    {
      "text": "File b03",
      "thumbnail": "",
      "attribute":4
    }
    {
      "text": "File b04",
      "thumbnail": "",
      "attribute":4
    }
    {
      "text": "File b05",
      "thumbnail": "",
      "attribute":4
    }
    {
      "text": "File b06",
      "thumbnail": "",
      "attribute":4
    }
    {
      "text": "File b07",
      "thumbnail": "",
      "attribute":4
    }
  ]
}
```

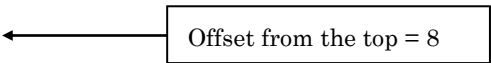
With this example, max_line is 10 so the remaining information more than line 8 is needed.
To do so, set “index = 8” in the Request Parameter.

Request

`http://{host}/YamahaExtendedControl/v1/netusb/getListInfo?input=usb&index=8&size=8&lang=en`

Response

```
{
  "response_code":0,
  "menu_layer":1,
  "max_line": 10
  "index":8,
  "playing_index":-1,
  "menu_name":"Folder B",
  "list_info":[
    {
      "text": "File b08",
      "thumbnail": "",
      "attribute":4
    }
  ]
}
```



```

    },
    {
        "text": "File b09",
        "thumbnail": "",
        "attribute": 4
    },
]
}

```

Capability bit[2]: Play bit equal to 1

13.1.5. Playback a file

Here shows how to specify and playback File b09. Confirm if the attribute of Response Parameter is playable. And the index of Request Parameter is 9.

Request

`http://{host}/YamahaExtendedControl/v1/netusb/setListControl?list_id=main&type=play&index=9`

Response

```

{
    "response_code": 0
}

```

The File b09 in Folder B is now playback.

13.1.6. Moving back one layer

Here shows an example of how to move back to one layer.

Request

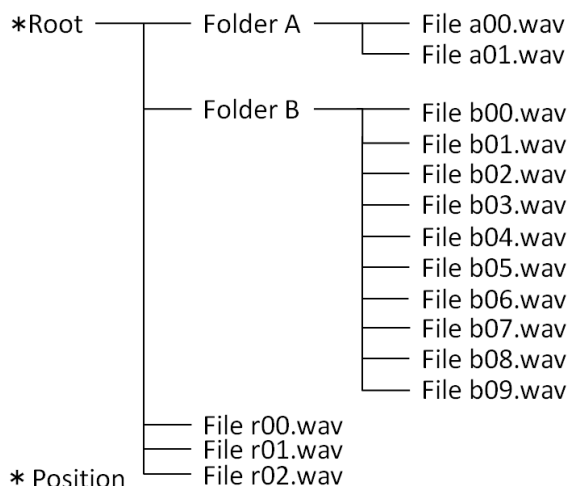
`http://{host}/YamahaExtendedControl/v1/netusb/setListControl
?list_id=main&type=return`

Response

```

{
    "response_code": 0
}

```



***Note**

getListInfo is the only command that blocks other commands execution. It may take up to 30 seconds to get the whole list information, and all other commands are not accepted all the while.

Retrieve list information via getListInfo after send setListControl:select command. It's possible to occur error if send setListControl:select command continuously.

○ : setListControl:select -> getList Info-> setListControl:select -> getListInfo

× : setListControl:select -> setListControl:select -> getListInfo

13.2. Device Search

Here explains procedures of how to search MusicCast Devices

1. Specify MediaRenderer and issue M-Search.
2. Access to the URL of device description read from the Location header in the response.
3. Confirm tags shown below.

<manufacturer>Yamaha Corporation</manufacturer>

See if manufacturer name is "Yamaha Corporation"

<yamaha:X_device>

See if there is this Yamaha tag

<yamaha:X_URLBase>http://192.168.10.103:80/</yamaha:X_URLBase>

Confirm device's IP address

<yamaha:X_yxcControlURL>/YamahaExtendedControl/v1/</yamaha:X_yxcControlURL>

Check the URL of YXC control

[Example of device description of a MusicCast Device]

```
<?xml version="1.0" encoding="utf-8" ?>
<root xmlns="urn:schemas-upnp-org:device-1-0"
xmlns:yamaha="urn:schemas-yamaha-com:device-1-0">
<specVersion>
  <major>1</major>
  <minor>0</minor>
</specVersion>
<device>
  <dlna:X_DLNAIDOC
xmlns:dlna="urn:schemas-dlna-org:device-1-0">DMR-1.50</dlna:X_DLNAIDOC>
  <deviceType>urn:schemas-upnp-org:device:MediaRenderer:1</deviceType>
  <friendlyName>Room A</friendlyName>
  <manufacturer>Yamaha Corporation</manufacturer>
  <manufacturerURL>http://www.yamaha.com/</manufacturerURL>
  <modelDescription>MusicCast</modelDescription>
```

```
<modelName>WXC-50</modelName>
<modelNumber>50</modelNumber>
<modelURL>http://www.yamaha.com/</modelURL>
<serialNumber>0DA27313</serialNumber>
<UDN>uuid:9ab0c000-f668-11de-9976-00a0ded26c17</UDN>
<iconList>
<icon>
  <mimetype>image/jpeg</mimetype>
  <width>48</width>
  <height>48</height>
  <depth>24</depth>
  <url>/Icons/48x48.jpg</url>
</icon>
<icon>
  <mimetype>image/jpeg</mimetype>
  <width>120</width>
  <height>120</height>
  <depth>24</depth>
  <url>/Icons/120x120.jpg</url>
</icon>
<icon>
  <mimetype>image/png</mimetype>
  <width>48</width>
  <height>48</height>
  <depth>24</depth>
  <url>/Icons/48x48.png</url>
</icon>
<icon>
  <mimetype>image/png</mimetype>
  <width>120</width>
  <height>120</height>
  <depth>24</depth>
  <url>/Icons/120x120.png</url>
</icon>
</iconList>
<serviceList>
<service>
  <serviceType>urn:schemas-upnp-org:service:AVTransport:1</serviceType>
  <serviceId>urn:upnp-org:serviceId:AVTransport</serviceId>
  <SCPDURL>/AVTransport/desc.xml</SCPDURL>
  <controlURL>/AVTransport/ctrl</controlURL>
  <eventSubURL>/AVTransport/event</eventSubURL>
</service>
<service>
  <serviceType>urn:schemas-upnp-org:service:RenderingControl:1</serviceType>
  <serviceId>urn:upnp-org:serviceId:RenderingControl</serviceId>
  <SCPDURL>/RenderingControl/desc.xml</SCPDURL>
  <controlURL>/RenderingControl/ctrl</controlURL>
  <eventSubURL>/RenderingControl/event</eventSubURL>
```

```

</service>
<service>
  <serviceType>urn:schemas-upnp-org:service:ConnectionManager:1</serviceType>
  <serviceId>urn:upnp-org:serviceId:ConnectionManager</serviceId>
  <SCPDURL>/ConnectionManager/desc.xml</SCPDURL>
  <controlURL>/ConnectionManager/ctrl</controlURL>
  <eventSubURL>/ConnectionManager/event</eventSubURL>
</service>
</serviceList>
<presentationURL>http://192.168.10.103/</presentationURL>
</device>
<yamaha:X_device>
  <yamaha:X_URLBase>http://192.168.10.103:80/</yamaha:X_URLBase>
  <yamaha:X_serviceList>
  <yamaha:X_service>

<yamaha:X_specType>urn:schemas-yamaha-com:service:X_YamahaRemoteControl:1</yamaha:X
_specType>
  <yamaha:X_controlURL>/YamahaRemoteControl/ctrl</yamaha:X_controlURL>
  <yamaha:X_unitDescURL>/YamahaRemoteControl/desc.xml</yamaha:X_unitDescURL>
  </yamaha:X_service>
  <yamaha:X_service>

<yamaha:X_specType>urn:schemas-yamaha-com:service:X_YamahaExtendedControl:1</yamaha
:X_specType>
  <yamaha:X_yxcControlURL>/YamahaExtendedControl/v1/</yamaha:X_yxcControlURL>
  <yamaha:X_yxcVersion>0801</yamaha:X_yxcVersion>
  </yamaha:X_service>
  </yamaha:X_serviceList>
  </yamaha:X_device>
</root>

```

13.3. Event Process

Events are spread out as UDP unicast. Retrieve necessary info by polling because of UDP packet loss.

(Ex) getStatus, getPlayInfo

13.4. Stereo Pair

Establish Stereo Pair setting by operating Devices. Stereo Pair setting is established as following.



Confirm Stereo Pair status.

Request

http://{DeviceA}/ YamahaExtendedControl/v1/system/getStereoPairInfo

Response

```
{
  "response_code":0,
  "status":"master_left",
  "pair_info":{
    "alive":true,
    "ip_address":"192.168.0.11",
    "mac_address":"00A0DE012345"
  }
}
```

Request

http://{DeviceB}/ YamahaExtendedControl/v1/system/getStereoPairInfo

Response

```
{
  "response_code":0,
  "status":"slave_right",
  "pair_info":{
    "alive":true,
    "ip_address":"192.168.0.10",
    "mac_address":"00A0DE012346"
  }
}
```

Control sender(master) device. Receiver(Slave) device operate simultaneously with sender(master) device.

Request

http://{DeviceA}/ YamahaExtendedControl/v1/main/setPower?power=standby

Response

```
{
  "response_code":0,
}
```



13.5. MusicCast Surround / Stereo

Under the condition of "Multi channel distribution", it is a function to combine multiple devices into one play set.

MusicCast Surround / Stereo types are the following three patterns.

- Stereo Pair (L + R)
- Surround Pair (Multi Main + SL + SR, Multi Main + SL/SR)
- Subwoofer Pair (Multi Main + SW, L + R + SW, L/R + SW)

In "MusicCast surround / stereo", the role that can be set for each device (which role of the system can be played) is decided. The roles are as follows. The parentheses indicate which role of Sender(Master) / Receiver(Slave).

- Multi channel main (Sender(Master))
- Surround L or R (Receiver(Slave))
- Surround L/R (Receiver(Slave))
- L/R (Sender(Master))
- Stereo Pair L or R (Sender(Master)/Receiver(Slave))
- Subwoofer (Receiver(Slave))

"MusicCast Surround / Stereo" combination list

		Receiver(Slave)						
		Multi Main	Stereo pair L	Stereo pair R	SL/SR	SL	SR	SW
Sender (Master)	Multi Main	N/A	N/A	N/A	●	N/A	N/A	○*1
		N/A	N/A	N/A	N/A	●	●	○*1
		N/A	N/A	N/A	N/A	N/A	N/A	●*1
	Stereo Pair L	N/A	N/A	●	N/A	N/A	N/A	○*1
	Stereo Pair R	N/A	●	N/A	N/A	N/A	N/A	○*1
	L/R	N/A	N/A	N/A	N/A	N/A	N/A	●*1

●Device indispensable for combination / ○Device that can be combined

*1 Sender(Master) capability : In the case that "can have Subwoofer" is "Yes"

To check the role of each device in MusicCast surround / stereo, refer to "master_role / slave_role" of "mc_surround" from the distribution data of getFeatures.

Request (URI)

http://{Device A}/ YamahaExtendedControl/v1/system/getFeatures

Response

```
{
  :
  "distribution":{
    :
    "mc_surround":{
      "master_role":{
        "surround_pair":true,
        "stereo_pair":false,
```



```
        "subwoofer_pair":true
    },
    "slave_role":{
        "surround_pair_l_or_r":false,
        "surround_pair_lr":false,
        "subwoofer_pair":false
    }
    :
    }
}
```

13.6. Polling

Polling interval of special application "MusicCast CONTROLLER".(reference information)

1. Always every 10 seconds for all rooms.
2. Always every 5 seconds for the selected MusicCast device.
(Since information can be acquired by events, please adjust the polling interval to an appropriate value depending on the system to be used.)

Polling is necessary for "get command" which may be updated in real time.

1. For all MusicCast devices being detected
[getLocationInfo](#)
2. For MusicCast devices of the same location only
[getDistributionInfo](#) (Please refer to YXC_API_Spec_Advanced.pdf)
3. For each room at the same location
/(zone)/[getStatus](#)
/(type)/[getPlayInfo](#) (Please use properly according to the current input, eg net/usb, tuner, cd...)

13.7. Special procedure for Amazon Music

About the procedure when Amazon Music is "Device Conflict (Account Conflict)". (reference information)

1. After a playback operation such as selection from the list, [netusb-play error:12](#) (Device Conflicted) is notified in the event of the YXC command.
2. Use [getPlayErrorInfo](#) to get the display content and display whether to play on the device. (Yes or No)
3. If "Yes" is selected, send "resume" with [setPlayback](#).
4. If "No" is selected, nothing is sent.

Revision History

Rev.	Date	Contents	Author
1.00	7/7/2016	First edition	Eto
1.10	9/28/2016	Expand control function. (Refer Revision_Modification.pdf for more detail)	Eto
1.20	1/26/2017	Add Stereo Pair control function. Add TIDAL/Deezer control function. Add remote management control function. Add note: storePreset, list control, and event process. (Refer Revision_Modification(Basic).pdf for more detail)	Eto
1.30	11/8/2017	Add note: polling Delete: switchAccount	Fukuyama
2.00	5/11/2018	Expand control function. (Refer Revision_Modification.pdf for more detail)	Fukuyama
2.10	8/9/2018	Expand control function. (Refer Revision_Modification.pdf for more detail)	Fukuyama
2.20	5/15/2019	Expand control function. (Refer Revision_Modification.pdf for more detail)	Fukuyama
2.30	8/1/2019	Expand control function. (Refer Revision_Modification.pdf for more detail)	Fukuyama
2.40	10/13/2020	Add new Response Parameters. Edit Master-> Sender (Master), Slave-> Receiver (Slave). (Refer Revision_Modification.pdf for more detail)	Kim
2.50	5/17/2021	Add new Response Parameters Expand control function. (Refer Revision_Modification.pdf for more detail) Review of item status	Kawai